

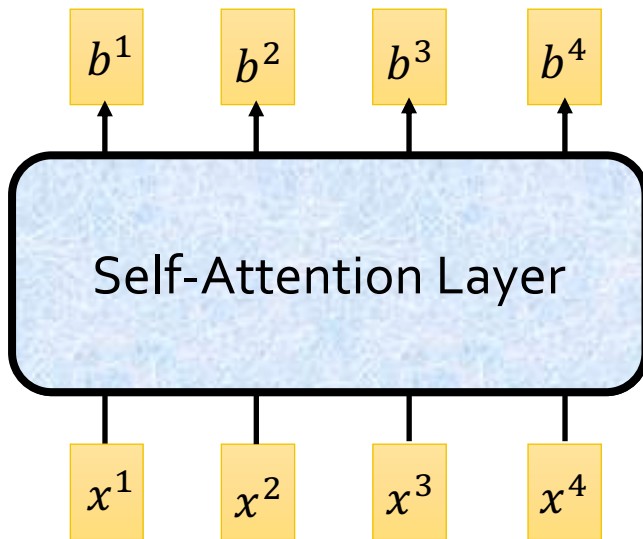
Transformer Architecture

CSCI 601-771 (NLP: Self-Supervised Models)

Self-Attention Module

Self-Attention a dimension-preserving function

- Given a set of input vector, compute output vectors of the same dimension.
- Now what's inside this box?



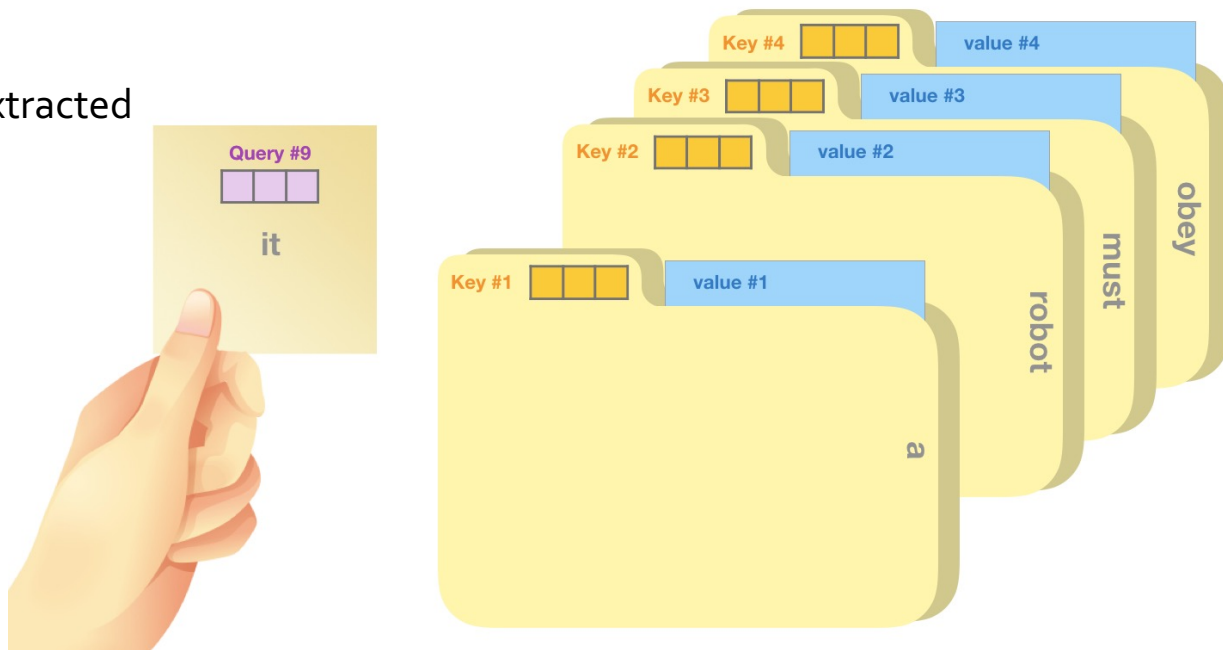
Defining Self-Attention

- Terminology:
 - **Query**: to match others
 - **Key**: to be matched
 - **Value**: information to be extracted

Defining Self-Attention

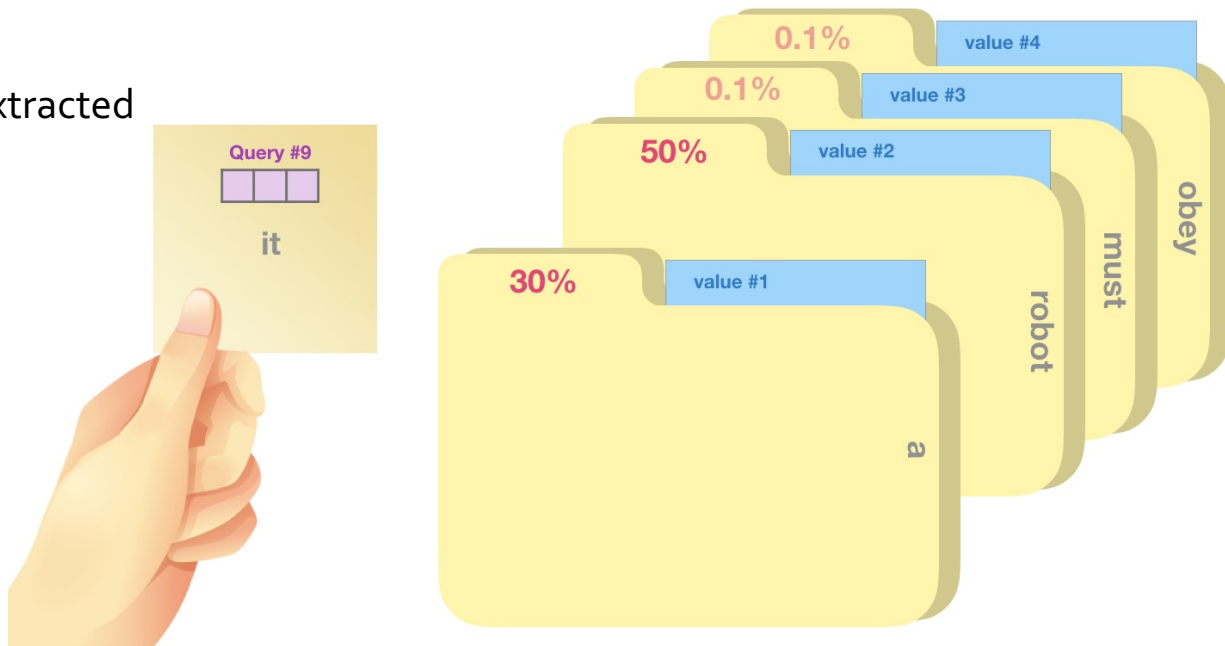
An analogy

- Terminology:
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 - **Value**: information to be extracted



Defining Self-Attention

- Terminology:
 - **Query**: to match others
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q : query (to match others)

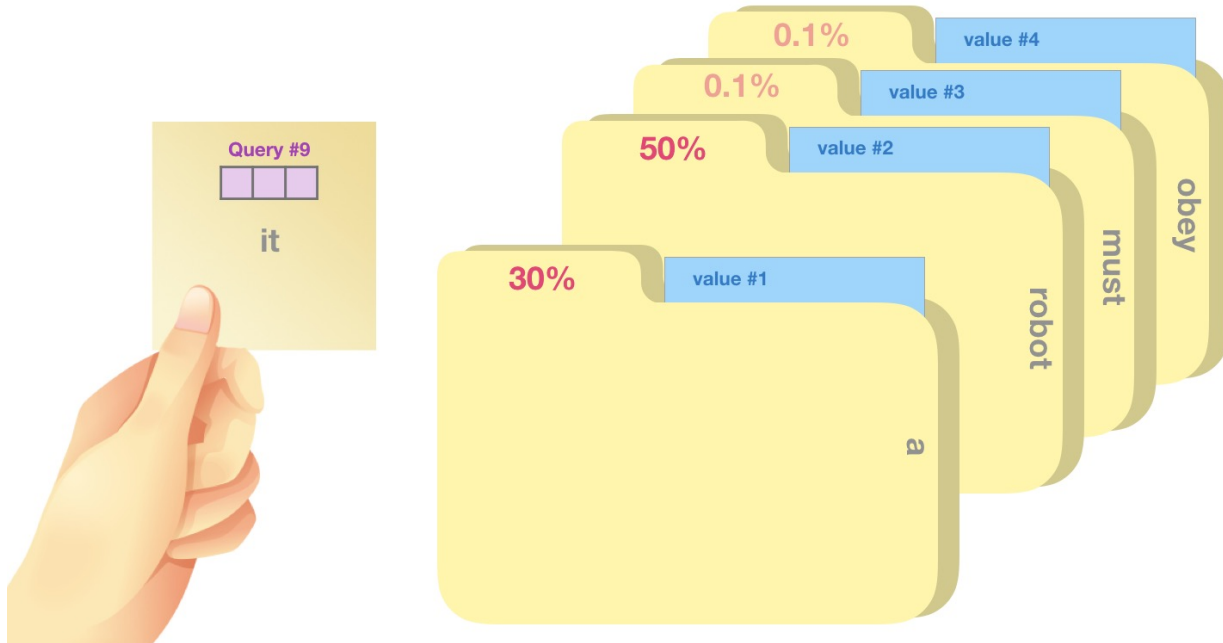
$$q_i = x_i W^q$$

k : key (to be matched)

$$k_i = x_i W^k$$

v : value (information to be extracted)

$$v_i = x_i W^v$$



q : query (to match others)

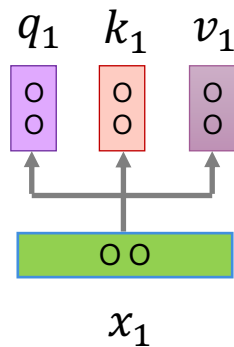
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The

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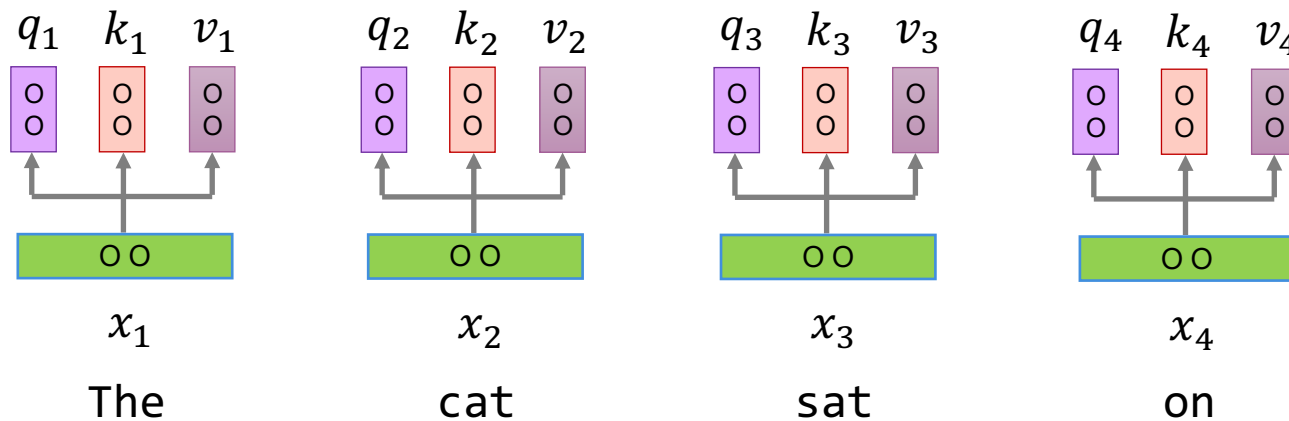
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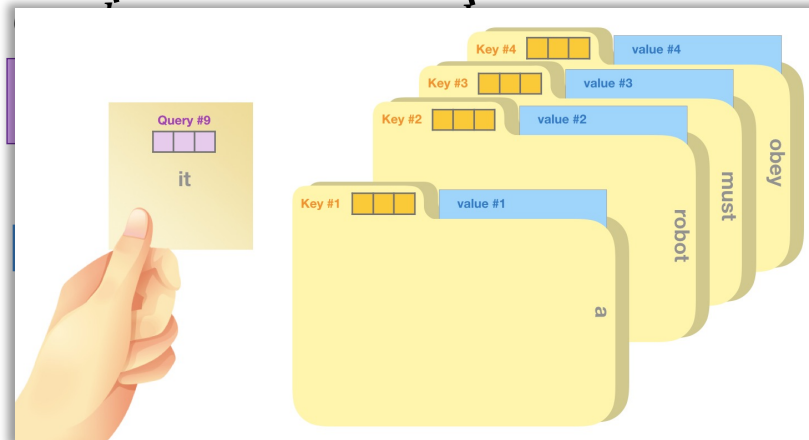
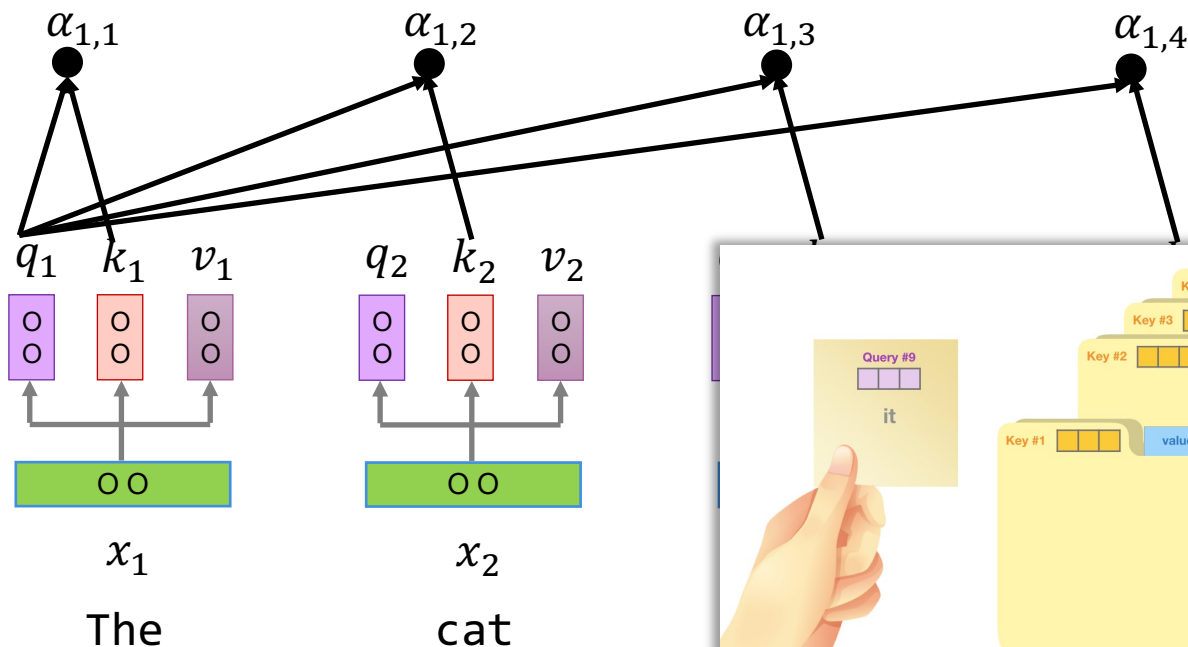
$$\alpha_{1,i} = \underbrace{q^1 \cdot k^i}_{\text{Scaled dot product}} / \sqrt{d}$$

q : query (to match others)

k : key (to be matched)

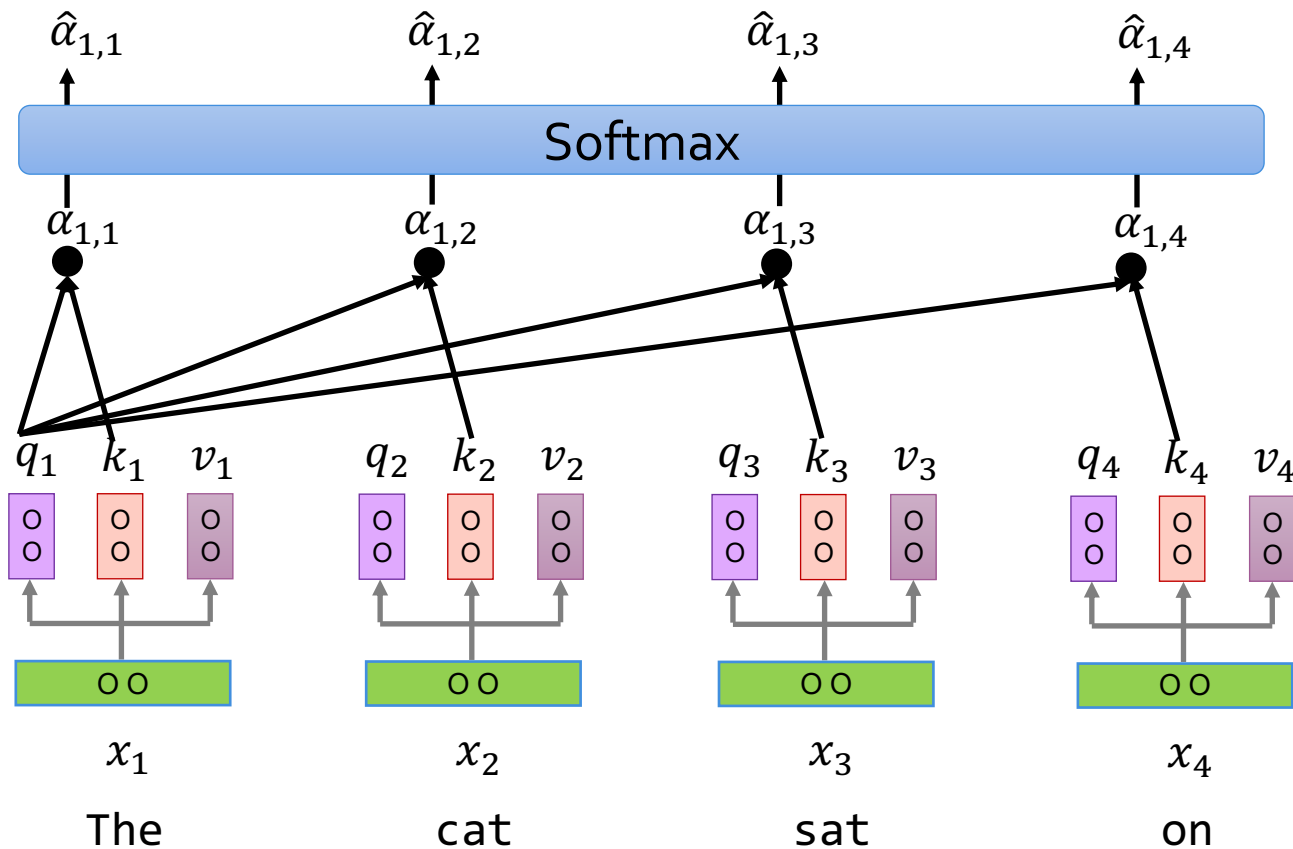
v : value (information to be extracted)

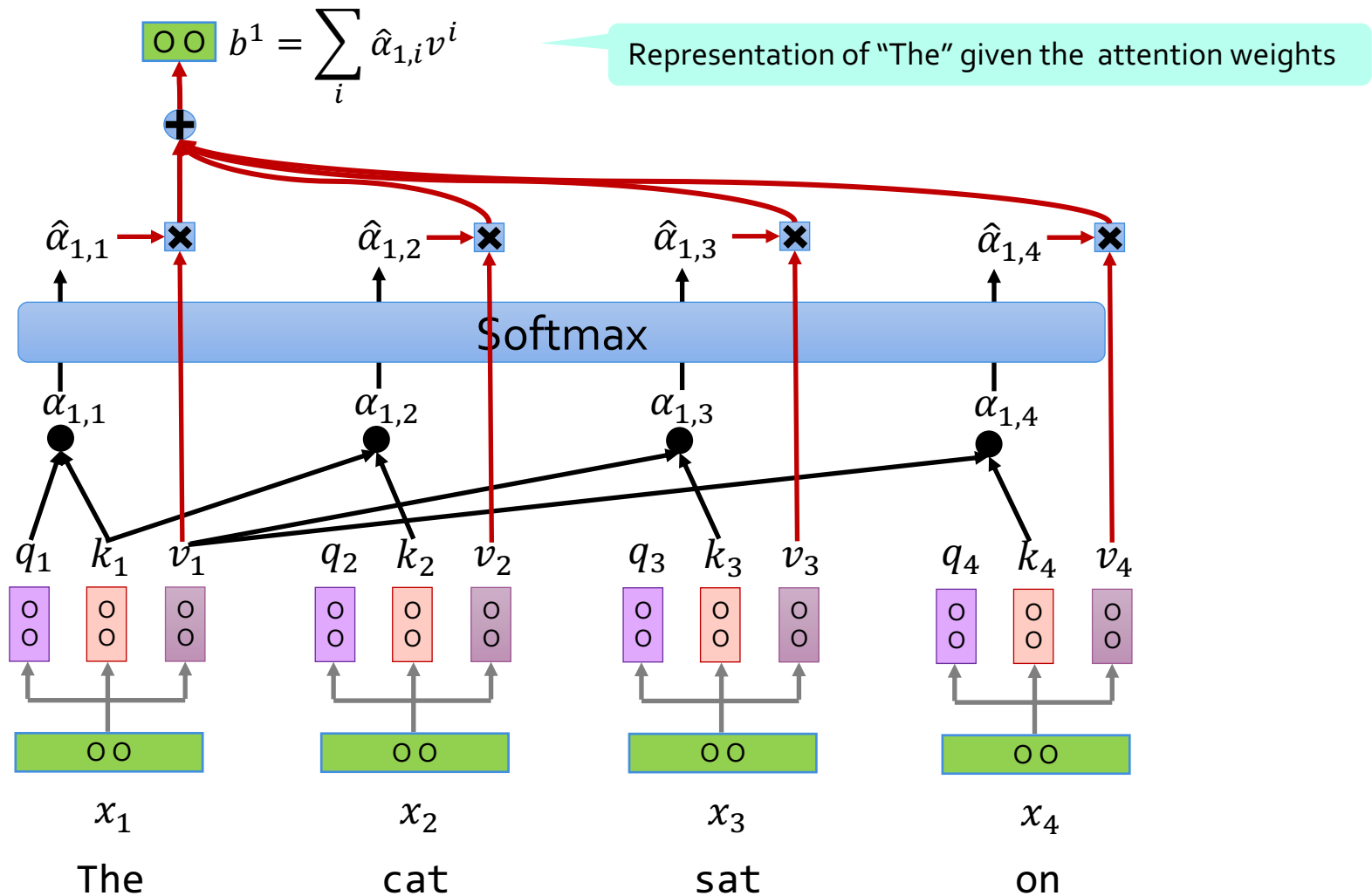
How much should "The" attend to other positions?



$$\sigma(z)_i = \frac{\exp(z_i)}{\sum_j \exp(z_j)}$$

How much
should "The"
attend to other
positions?





Self-Attention: Matrix Notation

- Input sequence: $\mathbf{x} = [x_1, x_2, \dots, x_n] \in \mathbb{R}^{n \times d}$

- Calculate:

$$\mathbf{Q} = \mathbf{x}\mathbf{W}^q \in \mathbb{R}^{n \times d}$$

$$\mathbf{K} = \mathbf{x}\mathbf{W}^k \in \mathbb{R}^{n \times d}$$

$$\mathbf{V} = \mathbf{x}\mathbf{W}^v \in \mathbb{R}^{n \times d}$$

- Pairwise similarity matrix between queries and keys: $\mathbf{Q}\mathbf{K}^T \in \mathbb{R}^{n \times n}$
- Attention output:

$$\text{Attention}(\mathbf{x}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right)\mathbf{V} \in \mathbb{R}^{n \times d}$$

Attention raw scores

0	-0.08	1.24	0.69	-0.98	1.43	-0.6	0.7	0.16	0.93	1.28	-1.61	-1.1
1	-0.09	-0.0	-0.7	0.06	0.25	0.23	0.26	0.18	0.78	-0.21	-1.01	1.01
2	0.86	1.19	1.59	0.86	-0.13	-0.15	-2.13	-0.98	-0.87	-1.72	1.87	-0.72
3	0.12	-0.03	-0.02	0.88	-0.46	-0.7	0.54	-0.42	-1.89	-0.38	0.04	-0.84
4	0.51	0.17	0.13	-1.64	0.24	-0.02	1.68	-0.36	0.64	0.36	0.27	0.66
5	0.24	-1.44	0.43	0.74	0.96	-1.21	-0.31	1.54	1.66	1.14	0.58	-1.44
6	0.26	-0.1	0.93	0.72	-0.38	1.65	0.47	-0.96	-0.17	-0.9	-1.57	0.22
7	-0.55	0.81	0.71	1.7	-0.8	-1.14	-0.32	1.78	-0.7	-0.04	1.54	0.81
8	0.74	-0.76	-0.44	-0.08	-1.38	-0.13	1.25	-1.37	1.84	0.3	0.57	0.74
9	-0.97	-0.91	0.15	0.35	-0.81	0.11	1.14	-1.52	1.06	1.87	0.5	-0.3
10	1.56	0.9	0.39	1.46	1.44	-1.05	0.9	-0.73	0.36	-0.67	-0.62	-0.43
11	0.32	0.74	0.44	-0.1	1.19	0.83	0.29	2.06	0.51	-0.26	1.51	0.11
	1	2	3	4	5	6	7	8	9	10	11	12

$$\begin{array}{c}
 \begin{array}{cc} \mathbf{Q} & \mathbf{K}^T \end{array} \\
 \begin{array}{c} \text{softmax}\left(\frac{\begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} \times \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array}}{\sqrt{d_k}}\right) \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} \end{array} \\
 \mathbf{V} \\
 \begin{array}{c} \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} \\ \hline \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} \end{array}
 \end{array}$$

Q: Why do we need $\sqrt{d}$?

$$\text{Attention}(\mathbf{x}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right)\mathbf{V}$$

- Without $\sqrt{d}$, the results of $\mathbf{Q}\mathbf{K}^T$ can be very large.
- Large differences between scores make softmax nearly one-hot
 - And consequently, small gradients during backprop.
- A theoretical justification: one can prove that:
 - if each element of $\mathbf{Q}$ and $\mathbf{K}$ are drawn independently from a $N(0, \sigma^2)$, then $\mathbf{Q}\mathbf{K}^T$ would have a variance $d\sigma^4$.
 - But $\mathbf{Q}\mathbf{K}^T/\sqrt{d}$ has variance of σ^4 (i.e., **independent of d**).
- This scaling prevents score variance from growing with dimension.

Q: Does this also handle batching?

$$\text{Attention}(\mathbf{x}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right)\mathbf{V}$$

Self-Attention: Thinking about batching

- Suppose our data comes in batches of size b , i.e., $\mathbf{x} \in \mathbb{R}^{b \times n \times d}$
- Self-attention's matrix form extends to this batched data. Let's verify:
- Calculate the keys, queries and values, per-batch:

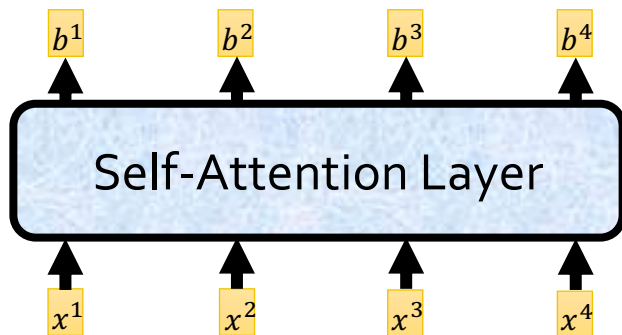
$$\begin{aligned}\mathbf{Q} &= \mathbf{x}\mathbf{W}^q \in \mathbb{R}^{b \times n \times d} \\ \mathbf{K} &= \mathbf{x}\mathbf{W}^k \in \mathbb{R}^{b \times n \times d} \\ \mathbf{V} &= \mathbf{x}\mathbf{W}^v \in \mathbb{R}^{b \times n \times d}\end{aligned}$$

- Pairwise similarity matrix between queries and keys, per batch: $\mathbf{Q}\mathbf{K}^T \in \mathbb{R}^{b \times n \times n}$
- Attention output for the whole batched data:

$$\text{Attention}(\mathbf{x}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right)\mathbf{V} \in \mathbb{R}^{b \times n \times d}$$

Recap: Self-Attention

- **Attention** is a powerful mechanism to create context-aware representations
- A way to focus on select parts of the input

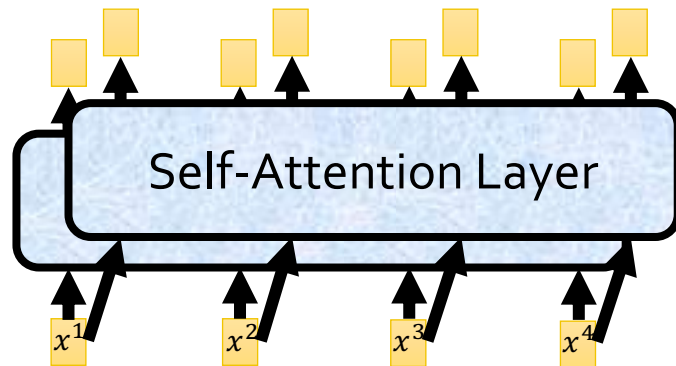


- Better at maintaining **long-distance dependencies** in the context.

Multi-Headed Self-Attention



- **Issue:** Each head has limited capacity.
 - A single head produces one weighted mixture per query; **multiple** heads can capture different relationships.
- **Main idea:** Allows model to jointly attend to information from different representation subspaces (like ensembling).
- **Multiple parallel attention layers.**
 - Each attention layer has its own parameters.
 - Concatenate the results and run them through a linear projection.

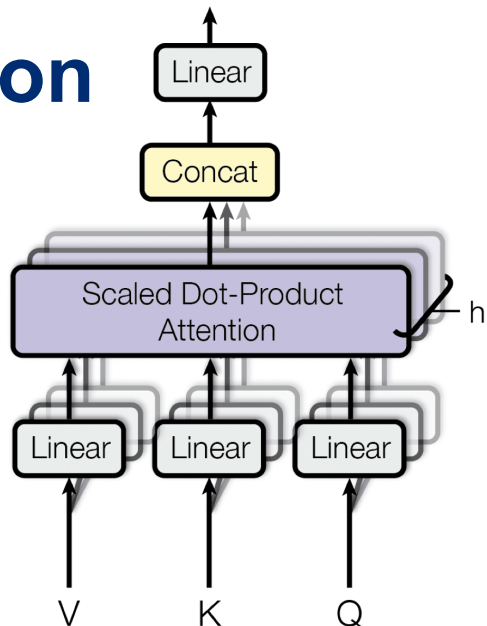


Multi-Headed Self-Attention

- Just concatenate all the heads and apply an output projection.

$$\text{head}_i = \text{Attention}(\mathbf{x}\mathbf{W}_i^q, \mathbf{x}\mathbf{W}_i^k, \mathbf{x}\mathbf{W}_i^v)$$
$$\text{MultiHeadedAttention}(\mathbf{x}) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)\mathbf{W}^o$$

- Q:** How do we change the dimensions of these parameters so that SA is dimension-preserving?



h : number of heads
 d : feature dimension
in output of SA

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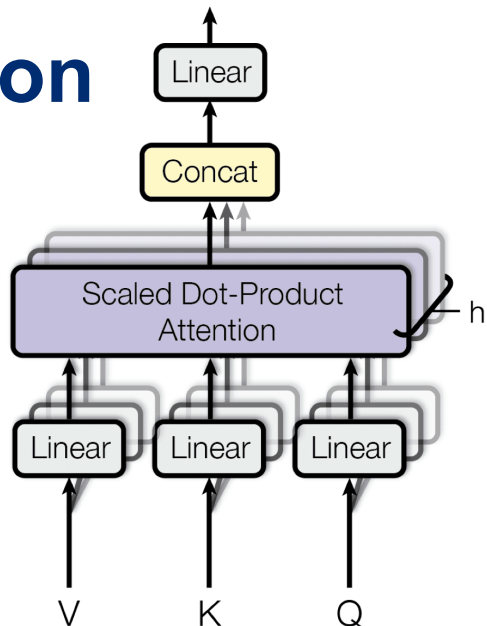
- Previously, we used the following dimensions for **single**-head SA:

$$\mathbf{W}_i^q \in \mathbb{R}^{d \times d}, \quad \mathbf{W}_i^k \in \mathbb{R}^{d \times d}, \quad \mathbf{W}_i^v \in \mathbb{R}^{d \times d},$$

- In practice, we use a reduced dimension for each head.

$$\mathbf{W}_i^q \in \mathbb{R}^{d \times \frac{d}{h}}, \quad \mathbf{W}_i^k \in \mathbb{R}^{d \times \frac{d}{h}}, \quad \mathbf{W}_i^v \in \mathbb{R}^{d \times \frac{d}{h}}, \quad \mathbf{W}^o \in \mathbb{R}^{d \times d}$$

- The total computational cost is similar to that of single-head attention with full dimensionality.



h : number of heads
 d : feature dimension
in output of SA

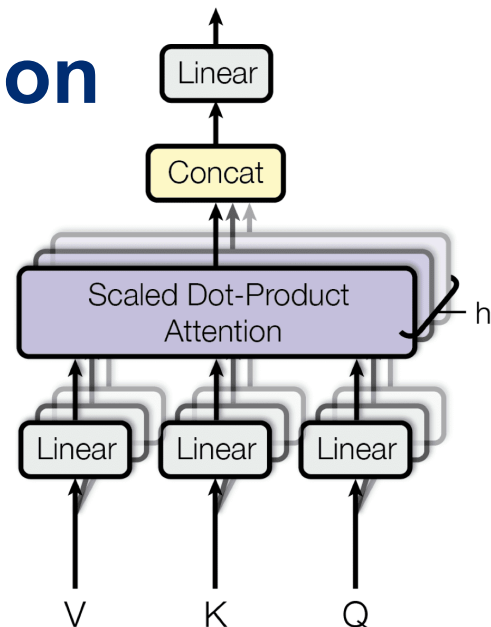
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$$\text{MultiHeadedAttention}(\mathbf{x}) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)\mathbf{W}^o$$

- Pulse check:**

- **Q1:** What is the rank of each head's output?
- **A1:** each head's outputs has rank at most $\min(n, d/h)$.
 - This is because the information pass through each head's rank low-dimensional bottleneck.
 - i.e., each head can only write within a d/h -dim subspace.
- **Q2:** What is the rank of the overall output of MHA?
- **A2:** Note the combined output (after concat and $\mathbf{W}^o$) can have rank up to $\min(n, d)$, where n is the sequence length.



h : number of heads
 d : feature dimension
in output of SA

Implementing Self-Attention Efficiently

- Modern LLMs don't use loops or separate W matrices. They instead use a fused QKV matrix.

```
def mha_textbook(x):  
    heads = []  
    for i in range(h):  
        q, k, v = x @ Wq[i], x @ Wk[i], x @ Wv[i]      # each (B, T, dh)  
        scores = q @ k.transpose(-2, -1) / math.sqrt(dh)  # (B, T, T)  
        scores = scores.masked_fill(mask, float("-inf"))  
        heads.append(torch.softmax(scores, dim=-1) @ v)    # (B, T, dh)  
    return torch.cat(heads, dim=-1) @ Wo                 # (B, T, d)
```

Given the fused weight ->

```
W_qkv = torch.cat([torch.cat(Wq, dim=1), # (d, d)
                    torch.cat(Wk, dim=1),
                    torch.cat(Wv, dim=1)], dim=1) # (d, 3d)
```

```
def mha_fused(x):
```

```
    qkv = x @ W_qkv
```

```
    # Split q/k/v and expose the head axis:
```

```
    # (B, T, 3d) -> (B, T, 3, h, dh) -> (3, B, h, T, dh)
```

```
    qkv = qkv.view(B, T, 3, h, dh).permute(2, 0, 3, 1, 4)
```

```
    q, k, v = qkv.unbind(0) # each (B, h, T, dh)
```

```
    # Batched over (B, h) -- the loop is gone, not reimplemented.
```

```
    scores = q @ k.transpose(-2, -1) / math.sqrt(dh) # (B, h, T, T)
```

```
    scores = scores.masked_fill(mask, float("-inf"))
```

```
    ctx = torch.softmax(scores, dim=-1) @ v # (B, h, T, dh)
```

```
    # The "Concat" box: undo the head split. Reshape only, zero FLOPs.
```

```
    ctx = ctx.transpose(1, 2).reshape(B, T, d)
```

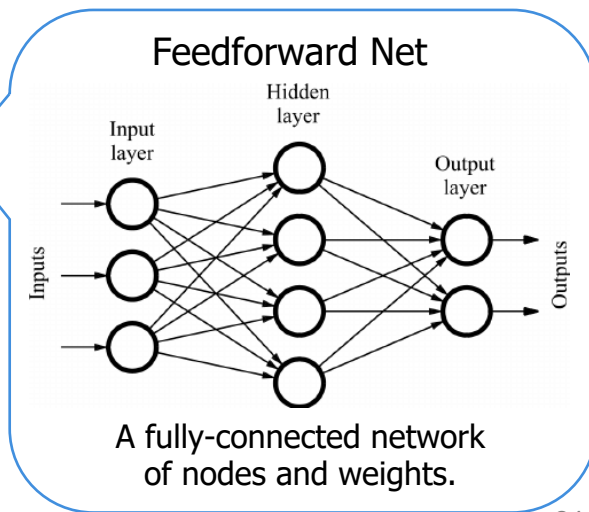
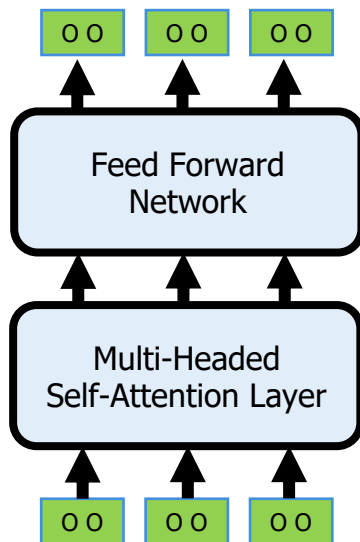
```
    return ctx @ Wo # ONE GEMM
```

Combine with FFN

- Add a **feed-forward network** to add more expressivity.
 - This applies another nonlinearity to the representations (or “post-process” them).
 - The same FFN is applied **independently** at every token position (unlike attention that mixes information between different positions).

$$\text{FFN}(\mathbf{x}) = f(\mathbf{x}\mathbf{W}_1 + b_1)\mathbf{W}_2 + b_2$$
$$\mathbf{W}_1 \in \mathbb{R}^{d \times d_{\text{ff}}},$$
$$\mathbf{W}_2 \in \mathbb{R}^{d_{\text{ff}} \times d}$$

- Usually, the dimensionality of the hidden feedforward layer d_{ff} is 2-8 times larger than the input dimension d .

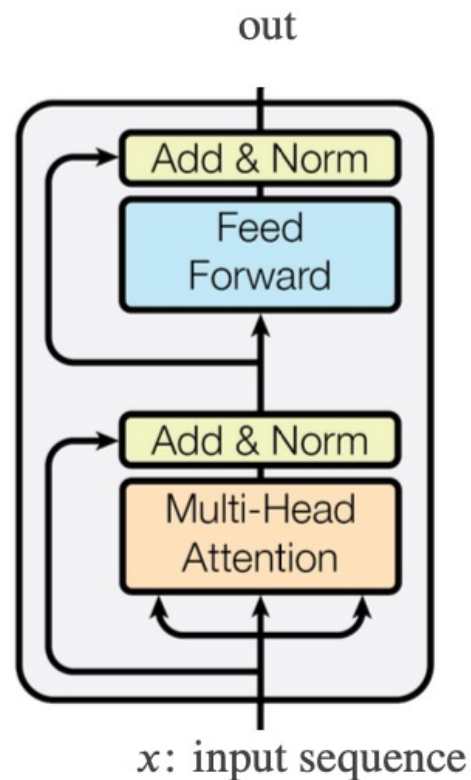


How Do We Prevent Vanishing Gradients?

- **Residual connections** let the model “skip” layers
 - These connections are particularly useful for training deep networks.

$$y = \frac{x - \mathbf{E}[x]}{\sqrt{\mathbf{Var}[x] + \epsilon}} * \gamma + \beta$$

- Use **layer normalization** to stabilize the network and allow for proper gradient flow.
- The visualized variant is called **post-norm**. Nowadays is more common to have them as **pre-norm**. (more on this later).



Transformer Block



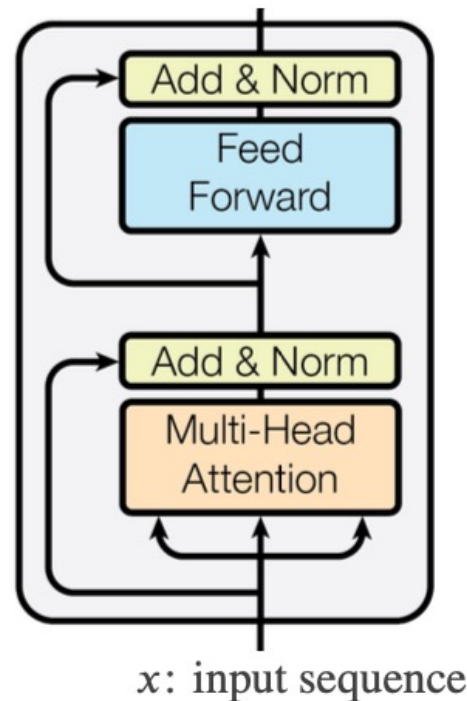
Given input $\mathbf{x}$:

- $\tilde{\mathbf{x}} = \text{MultiHeadedAttention}(\mathbf{x}; W^q, W^k, W^v, W^o)$

$$\mathbf{x} = \text{LayerNorm}(\tilde{\mathbf{x}} + \mathbf{x})$$

$$\tilde{\mathbf{x}} = \text{FFN}(\mathbf{x})$$

$$\text{out} = \text{LayerNorm}(\tilde{\mathbf{x}} + \mathbf{x})$$



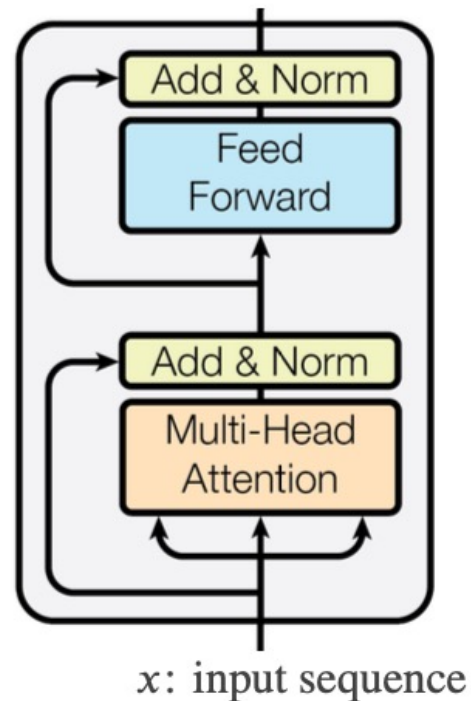
Pulse check

- **Q:** Each attention head uses one set of Q/K/V projection matrices shared across all token positions.
- **A: True.** Projections are shared across positions within a head.
- **Q:** Attention weights across queries sum to 1.
- **A: False.** Softmax is applied across **key positions**, separately for each query.

Summary: Transformer Block



- **Self-Attention:** A critical building block of modern language models.
 - The idea is to compose meanings of words weighted according some similarity notion.
- **Next:** We will extend this block to build various architectures known as Transformer.

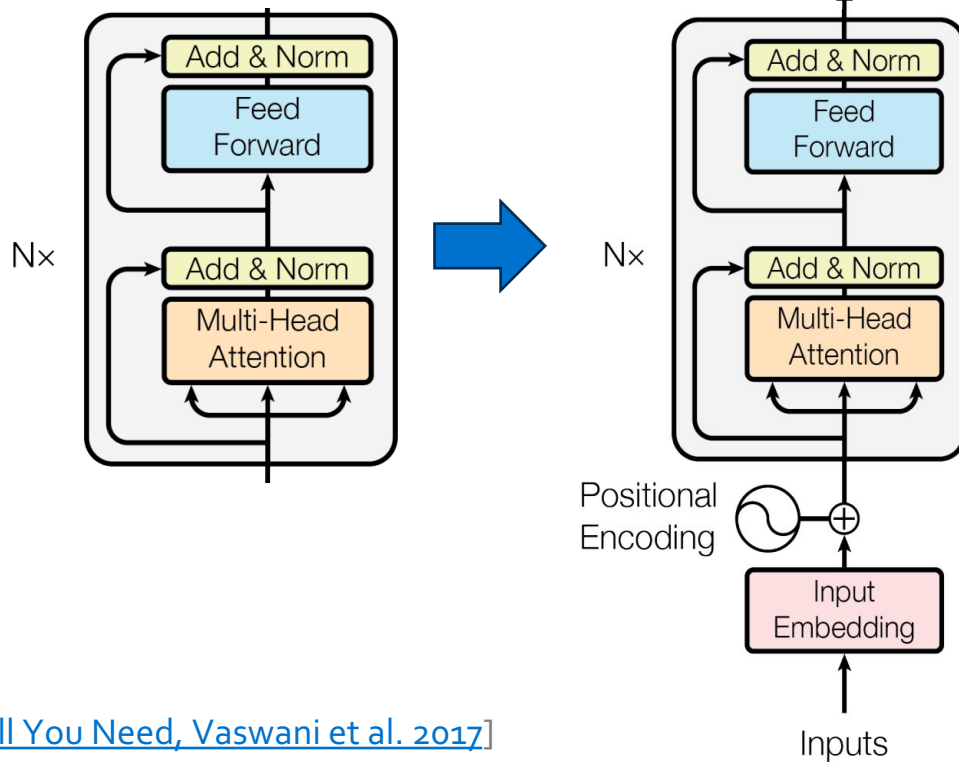
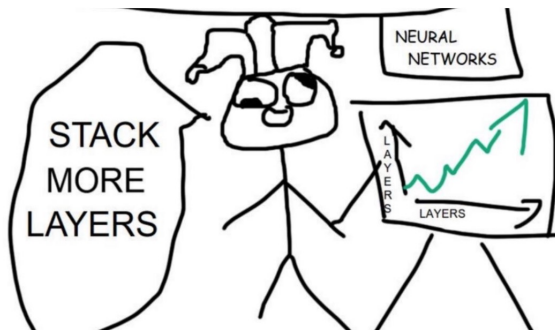




A Decoder-Only Transformer

How Do We Make it Deep?

- Stack more layers!



From Representations to Prediction

- To perform prediction, add a classification head on top of the final layer of the transformer: $\mathbf{x} \in \mathbb{R}^{n \times d}$.
- To obtain logits, we can apply a linear transformation with token embedding matrix $\mathbf{W}_V \in \mathbb{R}^{V \times d}$:

$$\mathbf{x}\mathbf{W}_V^T \in \mathbb{R}^{n \times V}$$

- To obtain probabilities, run this through softmax:

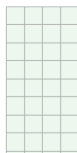
$$\text{softmax}(\mathbf{x}\mathbf{W}_V^T) \in \mathbb{R}^{n \times V}$$

Decoder #12, Position #1
output vector



X

Token
Embeddings



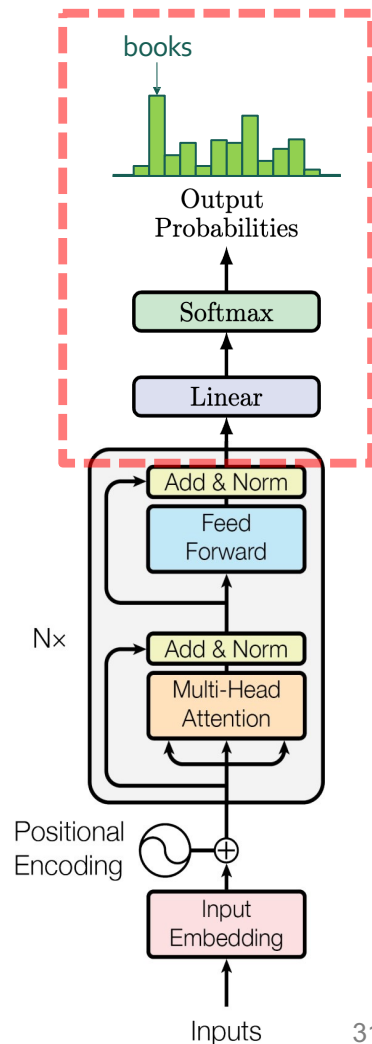
=

output token
probabilities (logits)

0.19850038	aardvark
0.7089803	aarhus
0.46333563	aaron
...	...
...	...
...	...
...	...
...	...

Pick an output
token based on
its probability
(sample)

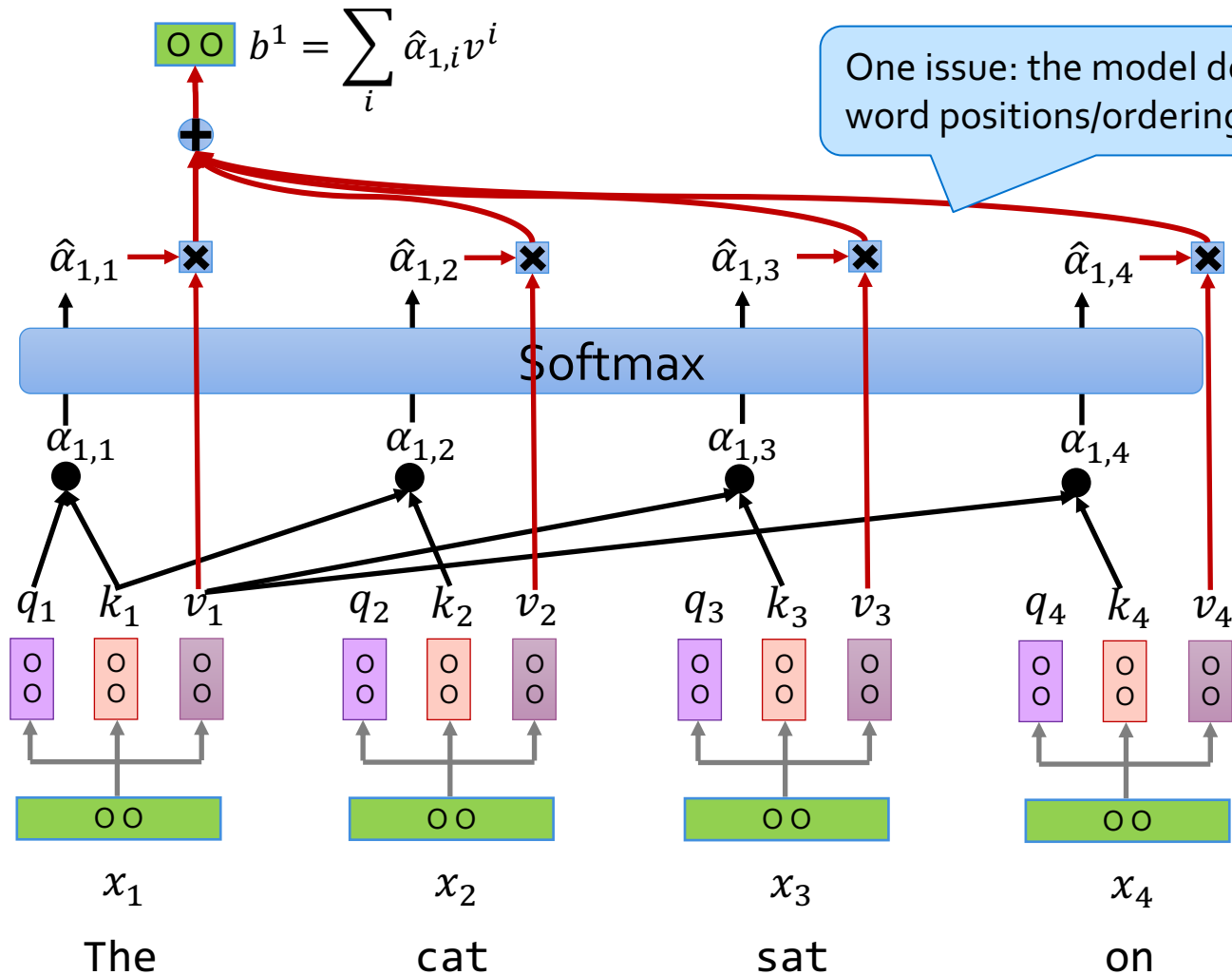
The





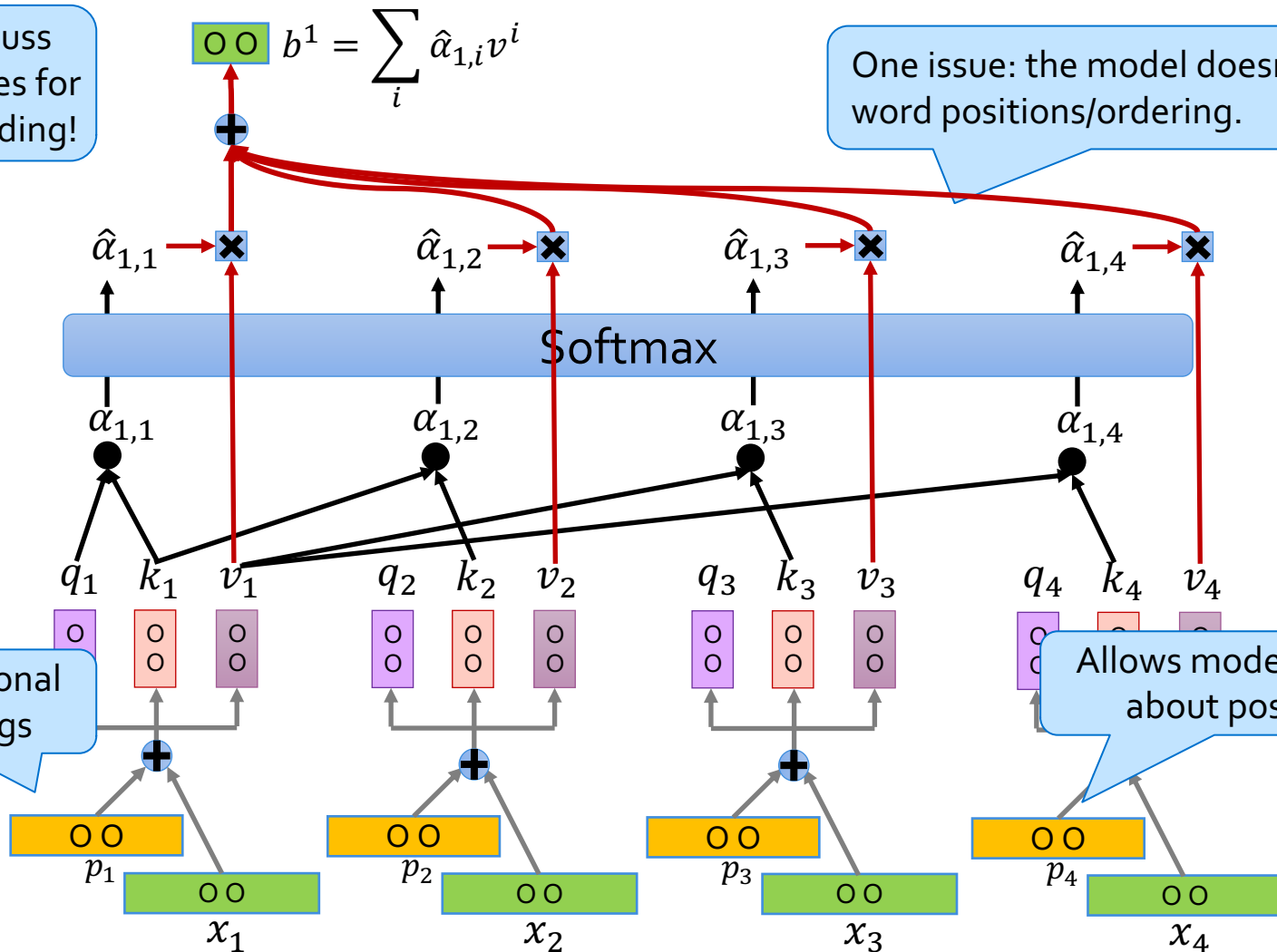
One wrinkle: How does Transformer
know about word ordering? ...





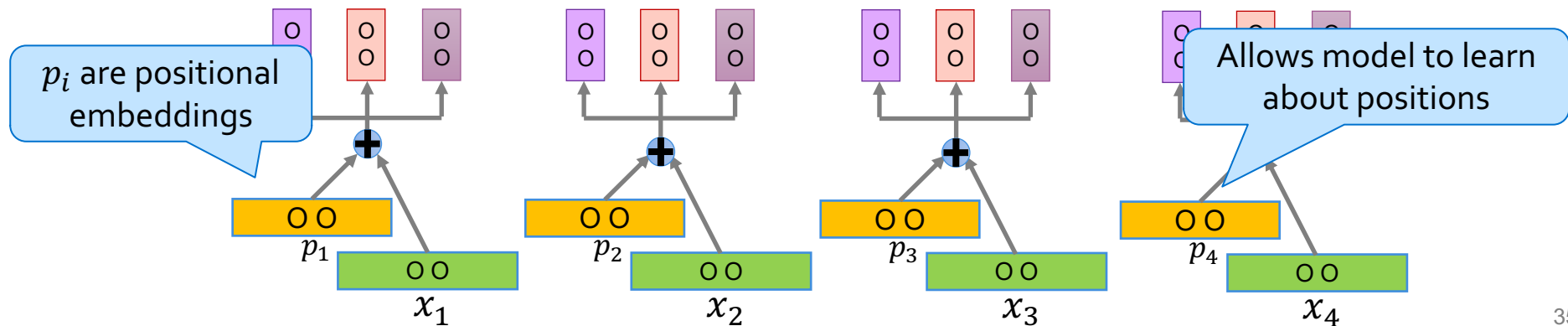
We will discuss various choices for these embedding!

One issue: the model doesn't know word positions/ordering.



Absolute Positional Embeddings

- Add unique vectors for each position.
- They inform Transformer about word orderings.
- We will see more variations of these in later class.

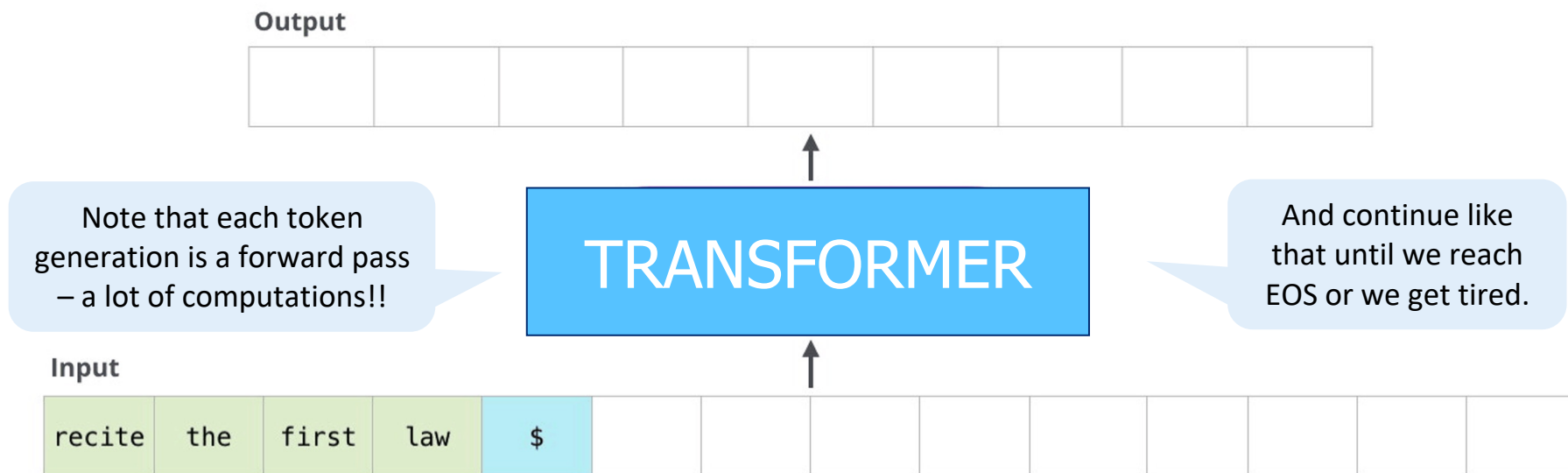




How does inference work?

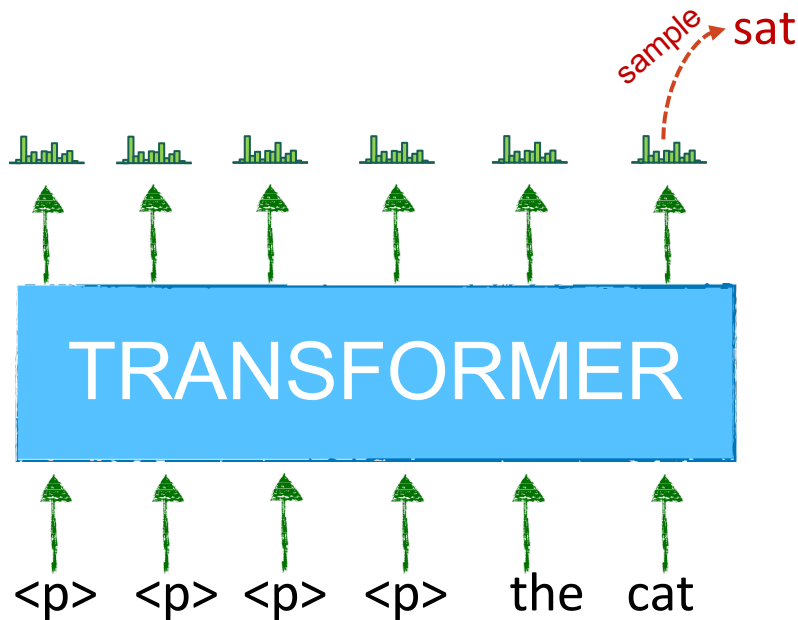


Generating text via Transformer



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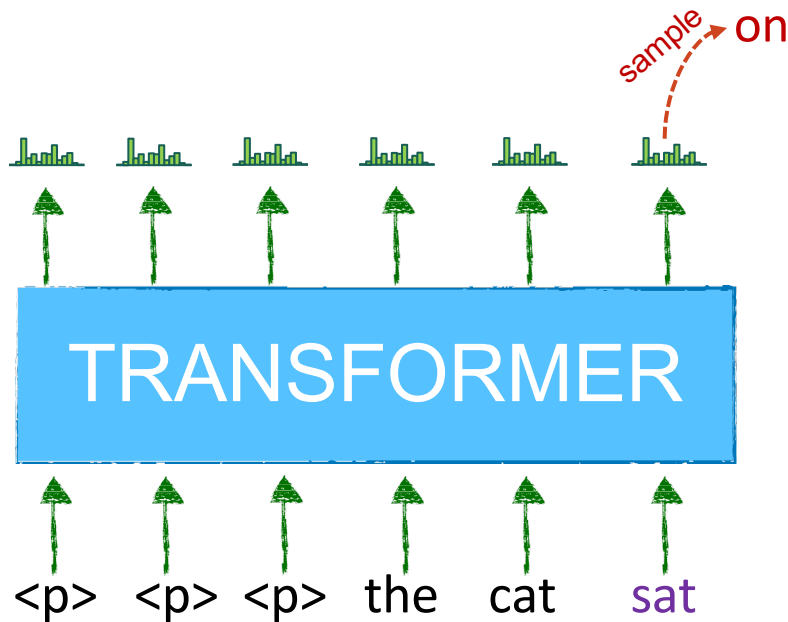
- Use the output of previous step as input to the next step repeatedly



Generating text via Transformer

- Use the output of previous step as input to the next step repeatedly

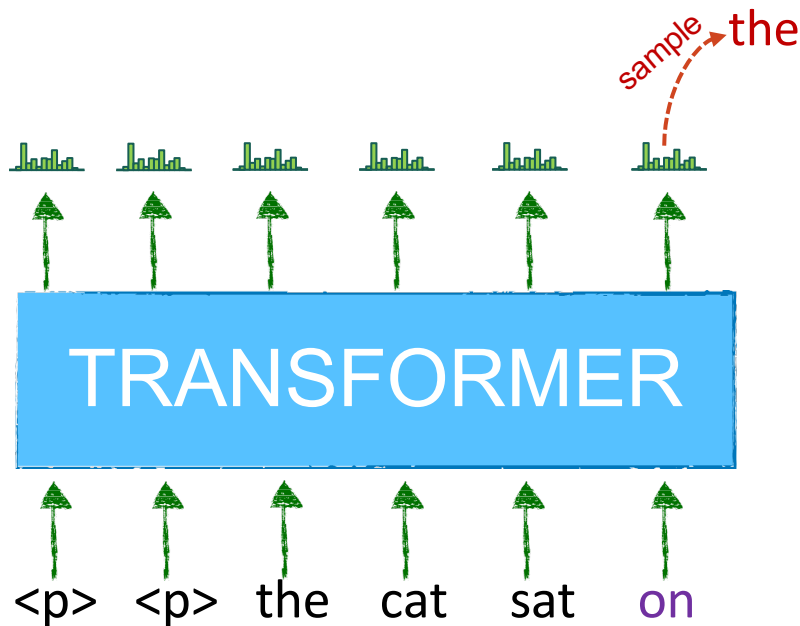
The probabilities get revised upon adding a new token to the input.



Generating text via Transformer

- Use the output of previous step as input to the next step repeatedly

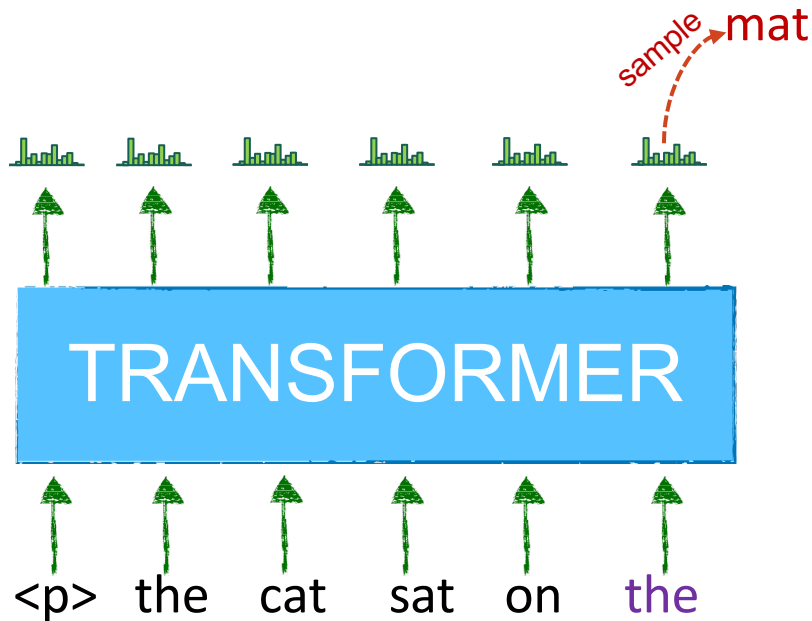
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Generating text via Transformer

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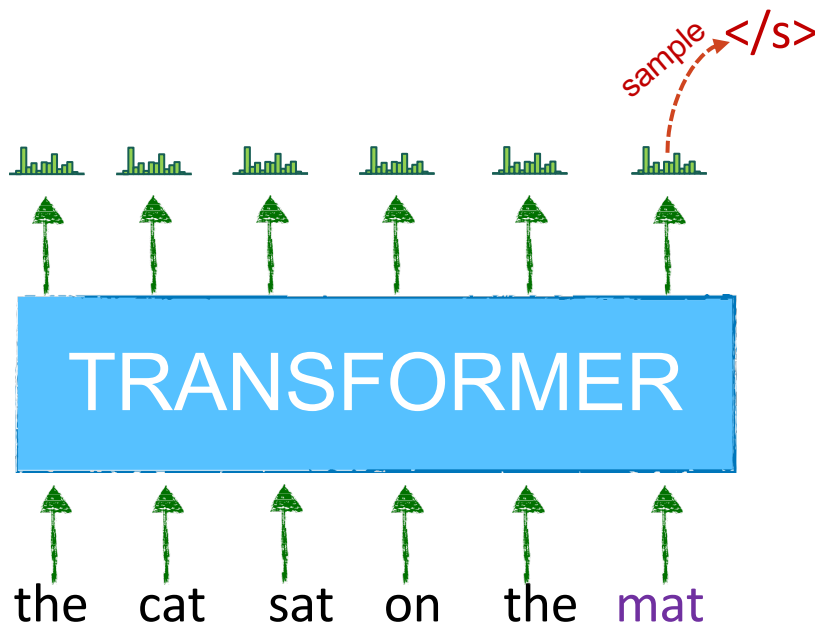
The probabilities get revised upon adding a new token to the input.



Generating text via Transformer

- Use the output of previous step as input to the next step repeatedly

The probabilities get revised upon adding a new token to the input.





How does training work?



Quiz

- What do we optimize when we train a transformer?
 1. X and q
 2. W_k and W_v
 3. Softmax function
 4. Sine/cos positional encodings
 5. The embedding matrix
 6. W_1 and W_2 in your FFN
 7. W_q

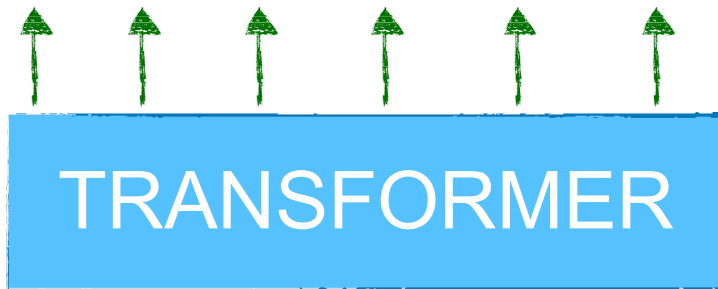
Answer: 2, 6, 7 and 5. You may also be able to train certain positional encodings.

Training a Transformer Language Model

- **Goal:** Train a Transformer for language modeling (i.e., predicting the next word).
- **Approach:** Train it so that each position is predictor of the next (right) token.
 - We just shift the input to left by one, and use as labels

(gold output) $Y =$ cat sat on the mat $</s>$

EOS special token



$X =$ the cat sat on the mat

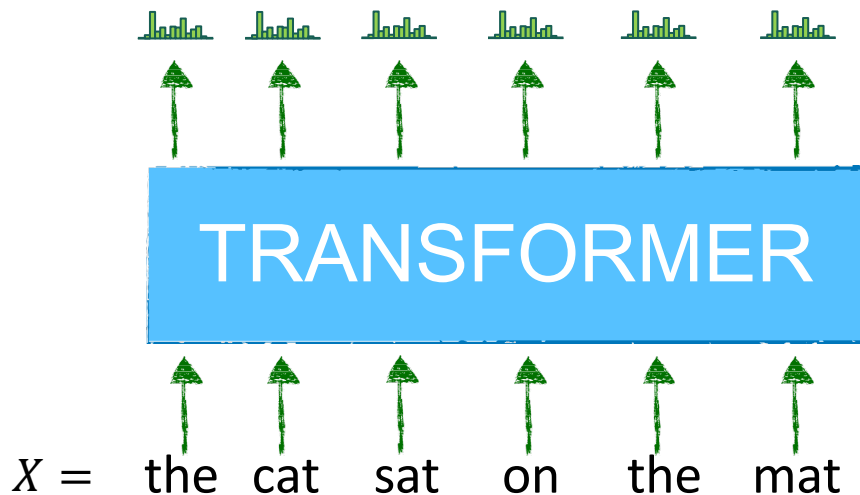
```
X = text[:, :-1]
Y = text[:, 1:]
```

[Slide credit: Arman Cohan]

Training a Transformer Language Model

- For each position, compute their corresponding **distribution** over the whole vocab.

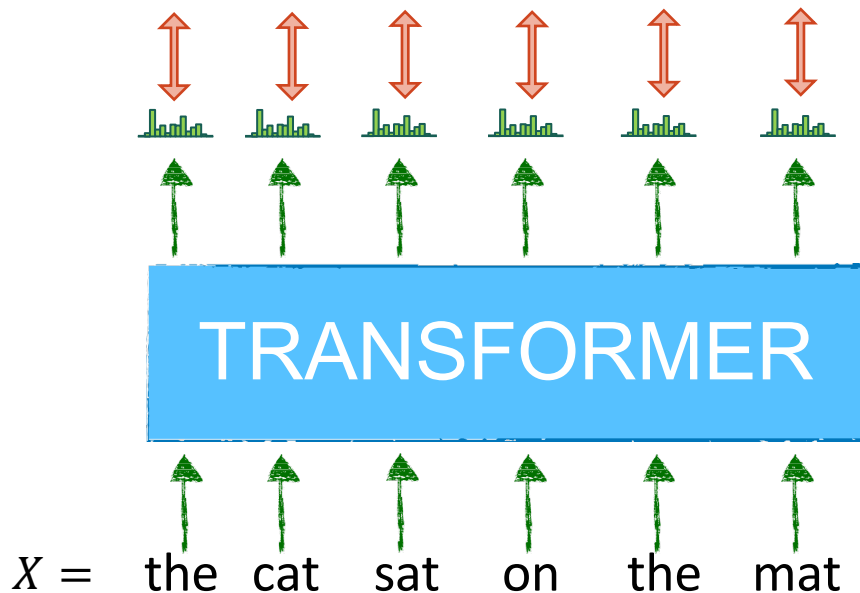
(gold output) $Y =$ cat sat on the mat $\langle /s \rangle$



Training a Transformer Language Model

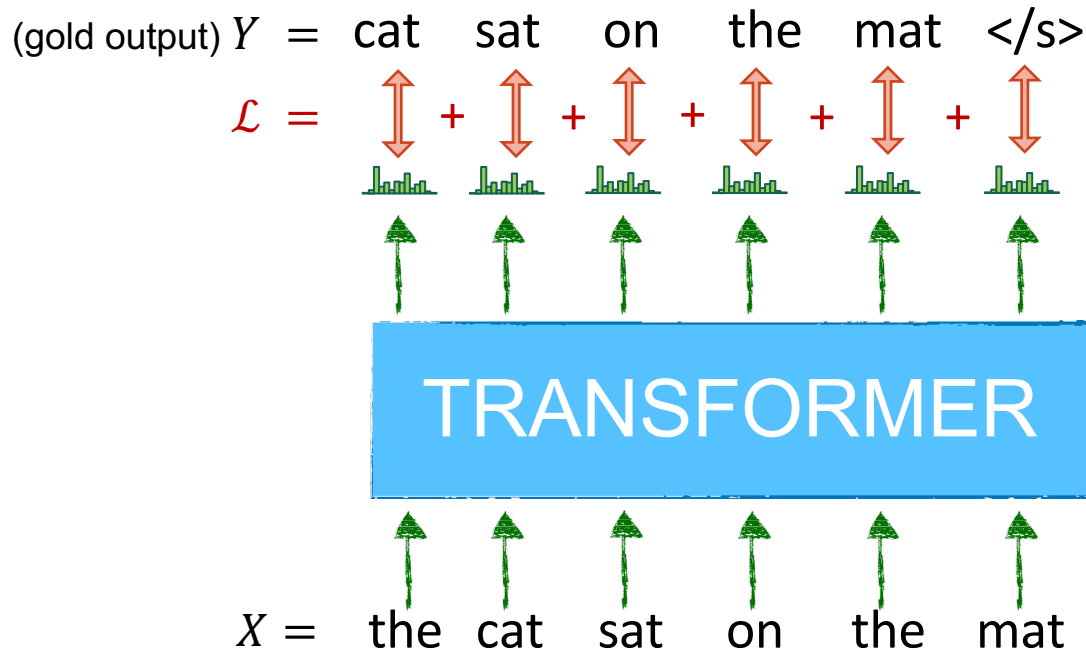
- For each position, compute the **loss** between the distribution and the gold output label.

(gold output) $Y =$ cat sat on the mat $</s>$



Training a Transformer Language Model

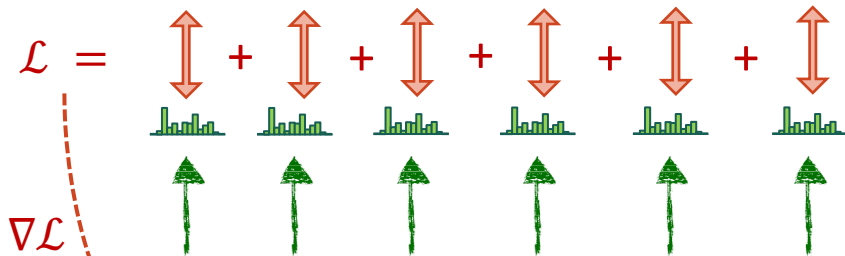
- Sum the position-wise loss values to obtain a **global loss**.



Training a Transformer Language Model

- Using this loss, do **Backprop** and **update** the Transformer parameters.

(gold output) $Y = \text{cat} \quad \text{sat} \quad \text{on} \quad \text{the} \quad \text{mat} \quad </s>$



TRANSFORMER

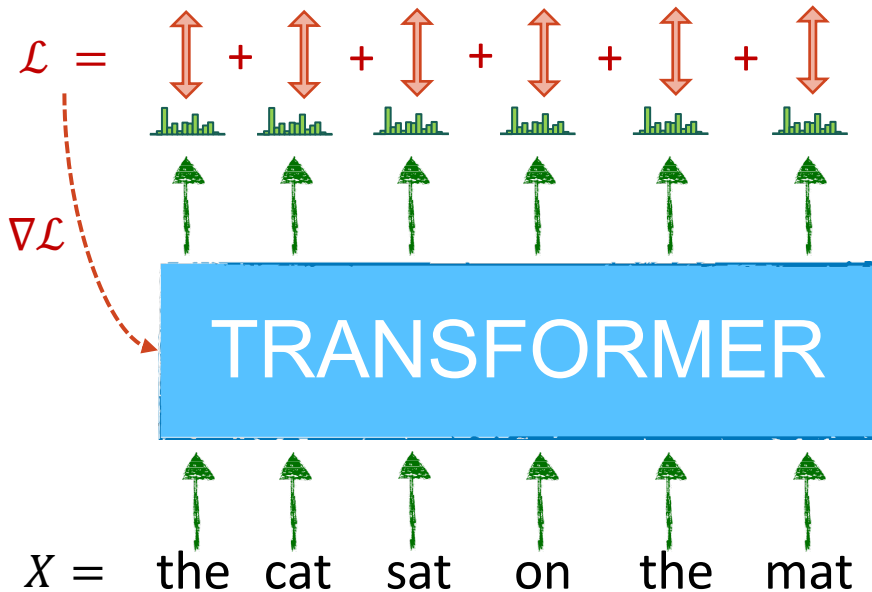
$X = \text{the} \quad \text{cat} \quad \text{sat} \quad \text{on} \quad \text{the} \quad \text{mat}$

Well, this is not quite right 🤔 ...
what is the problem with this?

Training a Transformer Language Model

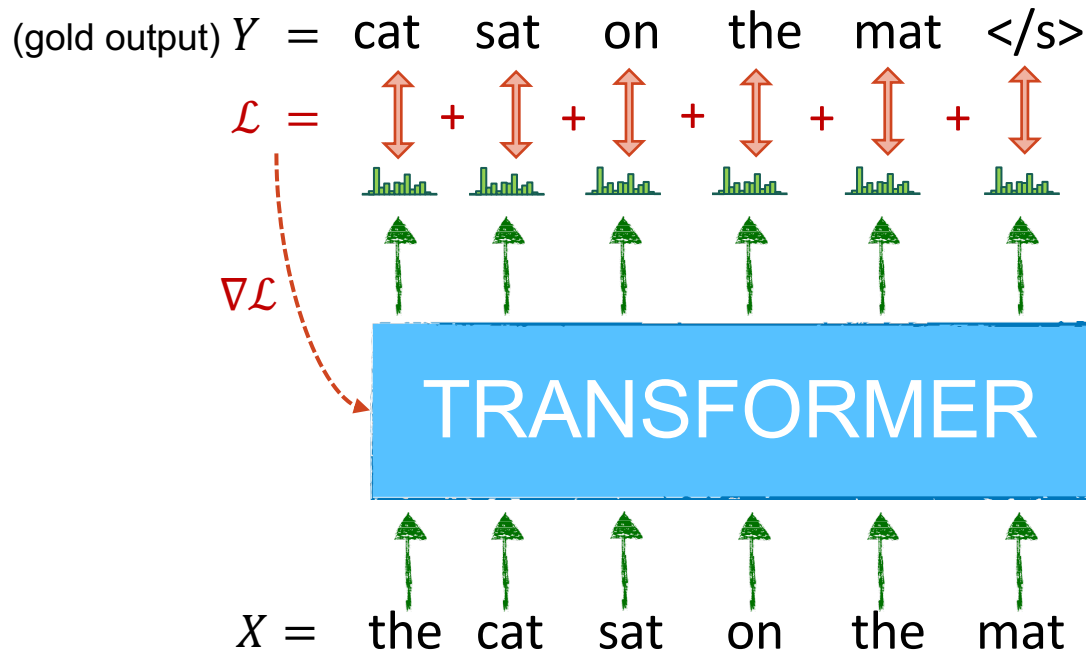
- The model would solve the task by **copying** the next token to output (data leakage).
 - Does **not** learn anything useful

(gold output) $Y =$ cat sat on the mat $</s>$



Training a Transformer Language Model

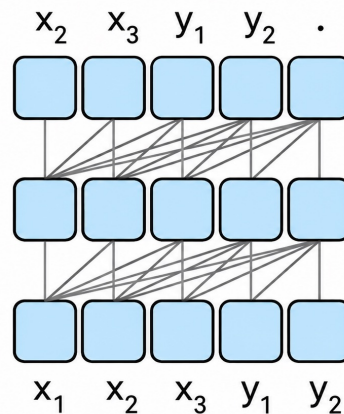
- We need to **prevent information leakage** from future tokens! How?



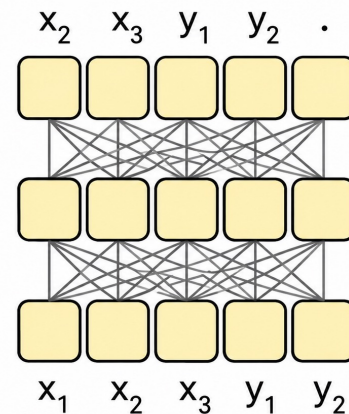
Attention mask

Attention raw scores

0	-0.08	1.24	0.69	-0.98	1.43	-0.6	0.7	0.16	0.93	1.28	-1.61	-1.1
1	-0.09	-0.0	-0.7	0.06	0.25	0.23	0.26	0.18	0.78	-0.21	-1.01	1.01
2	0.86	1.19	1.59	0.86	-0.13	-0.15	-2.13	-0.98	-0.87	-1.72	1.87	-0.72
3	0.12	-0.03	-0.02	0.88	-0.46	-0.7	0.54	-0.42	-1.89	-0.38	0.04	-0.84
4	0.51	0.17	0.13	-1.64	0.24	-0.02	1.68	-0.36	0.64	0.36	0.27	0.66
5	0.24	-1.44	0.43	0.74	0.96	-1.21	-0.31	1.54	1.66	1.14	0.58	-1.44
6	0.26	-0.1	0.93	0.72	-0.38	1.65	0.47	-0.96	-0.17	-0.9	-1.57	0.22
7	-0.55	0.81	0.71	1.7	-0.8	-1.14	-0.32	1.78	-0.7	-0.04	1.54	0.81
8	0.74	-0.76	-0.44	-0.08	-1.38	-0.13	1.25	-1.37	1.84	0.3	0.57	0.74
9	-0.97	-0.91	0.15	0.35	-0.81	0.11	1.14	-1.52	1.06	1.87	0.5	-0.3
10	1.56	0.9	0.39	1.46	1.44	-1.05	0.9	-0.73	0.36	-0.67	-0.62	-0.43
11	0.32	0.74	0.44	-0.1	1.19	0.83	0.29	2.06	0.51	-0.26	1.51	0.11
	1	2	3	4	5	6	7	8	9	10	11	12

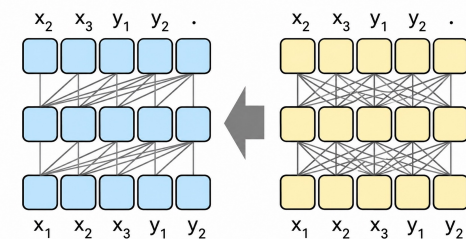


What we want



What we have

Attention mask



Attention raw scores

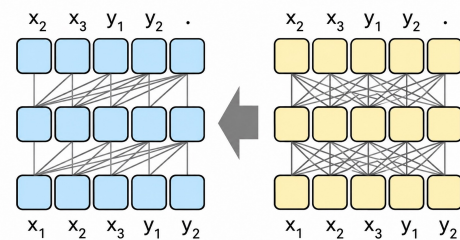
0	-0.08	1.24	0.69	-0.98	1.43	-0.6	0.7	0.16	0.93	1.28	-1.61	-1.1
1	-0.09	-0.0	-0.7	0.06	0.25	0.23	0.26	0.18	0.78	-0.21	-1.01	1.01
2	0.86	1.19	1.59	0.86	-0.13	-0.15	-2.13	-0.98	-0.87	-1.72	1.87	-0.72
3	0.12	-0.03	-0.02	0.88	-0.46	-0.7	0.54	-0.42	-1.89	-0.38	0.04	-0.84
4	0.51	0.17	0.13	-1.64	0.24	-0.02	1.68	-0.36	0.64	0.36	0.27	0.66
5	0.24	-1.44	0.43	0.74	0.96	-1.21	-0.31	1.54	1.66	1.14	0.58	-1.44
6	0.26	-0.1	0.93	0.72	-0.38	1.65	0.47	-0.96	-0.17	-0.9	-1.57	0.22
7	-0.55	0.81	0.71	1.7	-0.8	-1.14	-0.32	1.78	-0.7	-0.04	1.54	0.81
8	0.74	-0.76	-0.44	-0.08	-1.38	-0.13	1.25	-1.37	1.84	0.3	0.57	0.74
9	-0.97	-0.91	0.15	0.35	-0.81	0.11	1.14	-1.52	1.06	1.87	0.5	-0.3
10	1.56	0.9	0.39	1.46	1.44	-1.05	0.9	-0.73	0.36	-0.67	-0.62	-0.43
11	0.32	0.74	0.44	-0.1	1.19	0.83	0.29	2.06	0.51	-0.26	1.51	0.11
	1	2	3	4	5	6	7	8	9	10	11	12

Attention mask

0	0.0	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf
1	0.0	0.0	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf
2	0.0	0.0	0.0	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf
3	0.0	0.0	0.0	0.0	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf
4	0.0	0.0	0.0	0.0	0.0	-inf	-inf	-inf	-inf	-inf	-inf	-inf
5	0.0	0.0	0.0	0.0	0.0	0.0	-inf	-inf	-inf	-inf	-inf	-inf
6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-inf	-inf	-inf	-inf	-inf
7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-inf	-inf	-inf	-inf
8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-inf	-inf	-inf
9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-inf	-inf
10	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-inf
11	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	0	1	2	3	4	5	6	7	8	9	10	11

large negative numbers,
which leads to $\text{softmax}(-\infty) \approx 0$

Attention mask



Attention raw scores

0	-0.08	1.24	0.69	-0.98	1.43	-0.6	0.7	0.16	0.93	1.28	-1.61	-1.1
1	-0.09	-0.0	-0.7	0.06	0.25	0.23	0.26	0.18	0.78	-0.21	-1.01	1.01
2	0.86	1.19	1.59	0.86	-0.13	-0.15	-2.13	-0.98	-0.87	-1.72	1.87	-0.72
3	0.12	-0.03	-0.02	0.88	-0.46	-0.7	0.54	-0.42	-1.89	-0.38	0.04	-0.84
4	0.51	0.17	0.13	-1.64	0.24	-0.02	1.68	-0.36	0.64	0.36	0.27	0.66
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6	0.26	-0.1	0.93	0.72	-0.38	1.65	0.47	-0.96	-0.17	-0.9	-1.57	0.22
7	-0.55	0.81	0.71	1.7	-0.8	-1.14	-0.32	1.78	-0.7	-0.04	1.54	0.81
8	0.74	-0.76	-0.44	-0.08	-1.38	-0.13	1.25	-1.37	1.84	0.3	0.57	0.74
9	-0.97	-0.91	0.15	0.35	-0.81	0.11	1.14	-1.52	1.06	1.87	0.5	-0.3
10	1.56	0.9	0.39	1.46	1.44	-1.05	0.9	-0.73	0.36	-0.67	-0.62	-0.43
11	0.32	0.74	0.44	-0.1	1.19	0.83	0.29	2.06	0.51	-0.26	1.51	0.11
	1	2	3	4	5	6	7	8	9	10	11	12

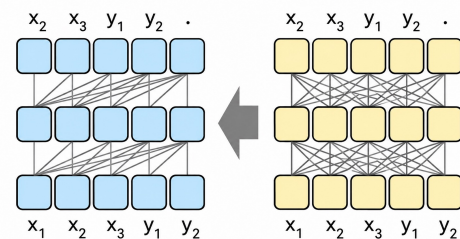
+

Apply
the mask
element-wise

Attention mask

0	0.0	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf
1	0.0	0.0	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf
2	0.0	0.0	0.0	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf
3	0.0	0.0	0.0	0.0	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf
4	0.0	0.0	0.0	0.0	0.0	-inf	-inf	-inf	-inf	-inf	-inf	-inf
5	0.0	0.0	0.0	0.0	0.0	0.0	-inf	-inf	-inf	-inf	-inf	-inf
6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-inf	-inf	-inf	-inf	-inf
7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-inf	-inf	-inf	-inf
8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-inf	-inf	-inf
9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-inf	-inf
10	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-inf
11	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	0	1	2	3	4	5	6	7	8	9	10	11

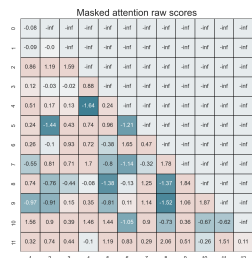
Attention mask



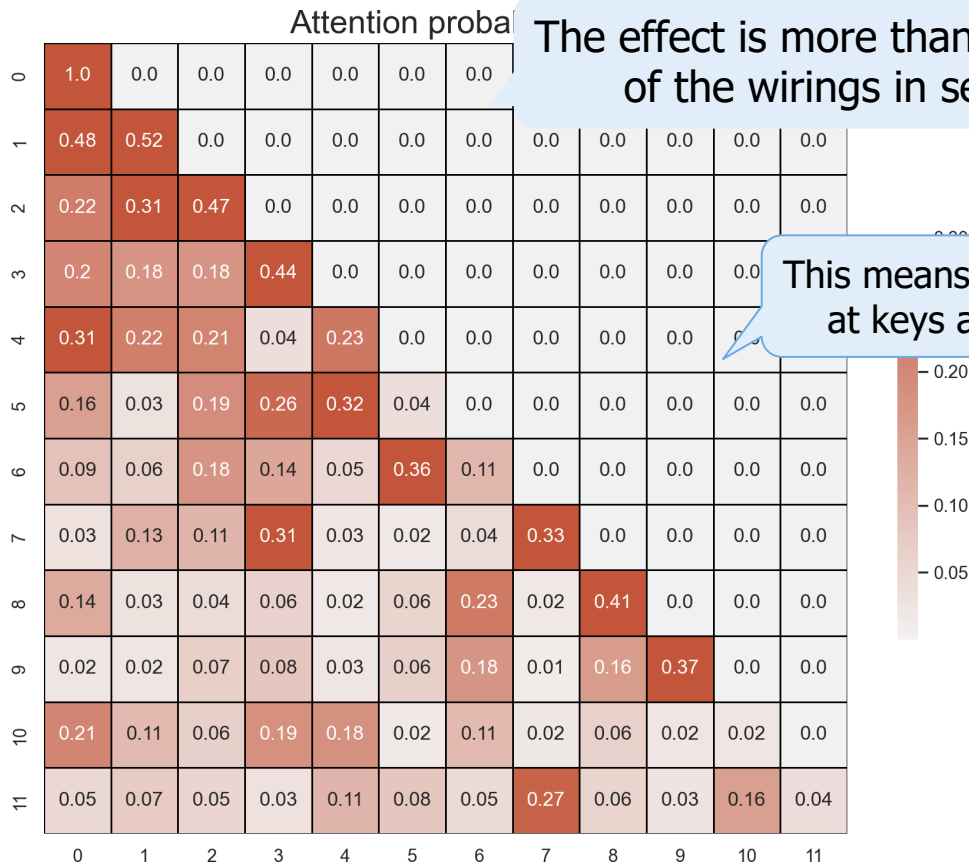
Masked attention raw scores

0	-0.08	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf
1	-0.09	-0.0	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf
2	0.86	1.19	1.59	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf
3	0.12	-0.03	-0.02	0.88	-inf	-inf	-inf	-inf	-inf	-inf	-inf
4	0.51	0.17	0.13	-1.64	0.24	-inf	-inf	-inf	-inf	-inf	-inf
5	0.24	-1.44	0.43	0.74	0.96	-1.21	-inf	-inf	-inf	-inf	-inf
6	0.26	-0.1	0.93	0.72	-0.38	1.65	0.47	-inf	-inf	-inf	-inf
7	-0.55	0.81	0.71	1.7	-0.8	-1.14	-0.32	1.78	-inf	-inf	-inf
8	0.74	-0.76	-0.44	-0.08	-1.38	-0.13	1.25	-1.37	1.84	-inf	-inf
9	-0.97	-0.91	0.15	0.35	-0.81	0.11	1.14	-1.52	1.06	1.87	-inf
10	1.56	0.9	0.39	1.46	1.44	-1.05	0.9	-0.73	0.36	-0.67	-0.62
11	0.32	0.74	0.44	-0.1	1.19	0.83	0.29	2.06	0.51	-0.26	1.51
	1	2	3	4	5	6	7	8	9	10	11

Attention mask



softmax



The effect is more than just pruning out some of the wirings in self-attention block.

This means that the query-3 can look at keys at positions {0, 1, 2, 3}.

Attention mask: a real example

```
scores = (q @ k.transpose(-2, -1)) / math.sqrt(q.size(-1))

# Mask entries above the diagonal.
mask = torch.triu(
    torch.ones_like(scores, dtype=torch.bool), diagonal=1
)
scores = scores.masked_fill(mask, float("-inf"))

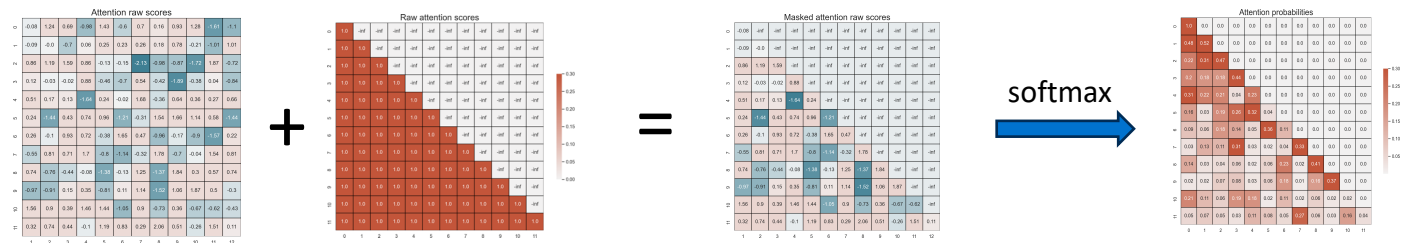
weights = torch.softmax(scores, dim=-1)
output = weights @ v
```

(2) The construction of the mask tensor is a bit different than our slides in practice.

(1) We're using a masking-specific function, not tensor addition. Why?

Attention masking: Why Before Softmax?

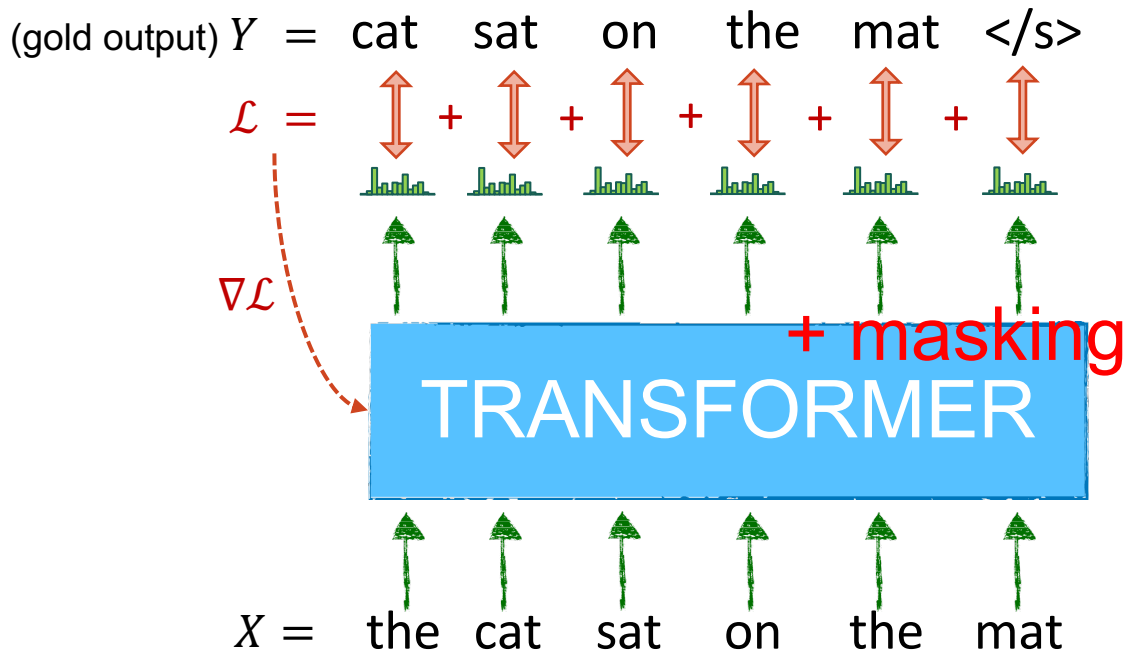
- We applied attention masking before softmax. Why not after?



- Softmax normalizes the scores so that it's a probability.
- Masking after softmax, would lead to an unnormalized probability distribution. So, you have to normalize twice! (wasted computation which we don't want)

Training a Transformer Language Model

- We need to **prevent information leakage** from future tokens! How?



Quiz: attention masking

- What's the point of attention masking during training?
 - A. speeds up training by reducing the number of computations performed.
 - B. prevents accessing future tokens, ensuring that predictions are based only on past and present information.
 - C. selectively highlights important tokens in the input, thereby enhancing attention mechanism.

Summary: A Decoder-only Transformer

- This is a very generic Transformer!
- **Next:** A more extended version of the Transformer architecture.

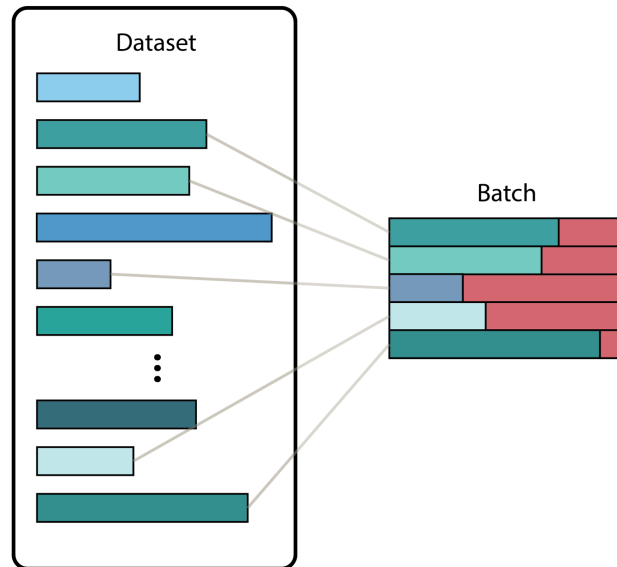
Padding sequences

- **Why pad?**

- GPUs process data in batches, i.e., tensor of shape (batch_size, seq_len).
- Sequences in a batch have different lengths;
- Padding makes their tensor shapes match.

- Padding is (often) done differently for training vs. inf.

- The two phases need different parts of the sequence.



Padding sequences

- **Inference uses left-padding:**

- Every prompt now ends in the same column
- The next token for the whole batch is read from the last position.
- We want to avoid scenario where various inputs (in the batch) need to be decoded from different dimensions.

- **Training uses right-padding:**

- Right padding is simple:
 - real content they all start at position 0 and labels line up with inputs.
- In theory, left-padding during training would also work
 - The model predicts every next token in parallel.
 - It doesn't matter where each sequence ends.

Inference · left padding

prompt (left-padded)

generated

input tokens

seq 0	The	cat	sat	on	the	mat
seq 1	[PAD]	[PAD]	I	love	deep	learning
seq 2	[PAD]	[PAD]	[PAD]	Hello	world	!
seq 3	[PAD]	Attention	is	all	you	need
	0	1	2	3	4	5
	column					

Training · right padding

input tokens

seq 0	The	cat	sat	on	the	mat	.
seq 1	I	love	deep	learning	[PAD]	[PAD]	[PAD]
seq 2	Attention	is	all	you	need	[PAD]	[PAD]
seq 3	Hello	world	[PAD]	[PAD]	[PAD]	[PAD]	[PAD]
	0	1	2	3	4	5	6
	column						

Padding sequences

- We also need to mask out padding tokens so we're not training for those PAD positions.
- Here `attention_mask` marks which tokens are real vs PAD.
- This is then applied to attention matrix:
 - The keys of PAD tokens are masked.
 - The pad queries can still attend to the real tokens, but their outputs are thrown away (ignored from the loss) since nothing requires pad queries to attend to anything.

Training · right padding

	input_ids						
seq 0	The	cat	sat	on	the	mat	.
seq 1	I	love	deep	learning	[PAD]	[PAD]	[PAD]
seq 2	Attention	is	all	you	need	[PAD]	[PAD]
seq 3	Hello	world	[PAD]	[PAD]	[PAD]	[PAD]	[PAD]
	0	1	2	3	4	5	6
	column						

	attention_mask (B, T)						
	1	1	1	1	1	1	1
	1	1	1	1	0	0	0
	1	1	1	1	1	0	0
	1	1	0	0	0	0	0
	0	1	2	3	4	5	6

seq 1: score mask (T, T)

	1	1	1	1	0	0	0
	↓	↓	↓	↓	↓	↓	↓
I	0.0	-inf	-inf	-inf	-inf	-inf	-inf
love	0.0	0.0	-inf	-inf	-inf	-inf	-inf
deep	0.0	0.0	0.0	-inf	-inf	-inf	-inf
learning	0.0	0.0	0.0	0.0	-inf	-inf	-inf
[PAD]	0.0	0.0	0.0	0.0	-inf	-inf	-inf
[PAD]	0.0	0.0	0.0	0.0	-inf	-inf	-inf
[PAD]	0.0	0.0	0.0	0.0	-inf	-inf	-inf
I	0.0	-inf	-inf	-inf	-inf	-inf	-inf
love	0.0	0.0	-inf	-inf	-inf	-inf	-inf
deep	0.0	0.0	0.0	-inf	-inf	-inf	-inf
learning	0.0	0.0	0.0	0.0	-inf	-inf	-inf
[PAD]	0.0	0.0	0.0	0.0	-inf	-inf	-inf
[PAD]	0.0	0.0	0.0	0.0	-inf	-inf	-inf
[PAD]	0.0	0.0	0.0	0.0	-inf	-inf	-inf

Padding sequences

- The logic is the same for inference: the keys of PAD tokens are masked since these tokens are uninformative.

Inference • left padding

prompt (left-padded)

generated

input_ids

seq 0	The	cat	sat	on	the	mat
seq 1	[PAD]	[PAD]	I	love	deep	learning
seq 2	[PAD]	[PAD]	[PAD]	Hello	world	!
seq 3	[PAD]	Attention	is	all	you	need
	0	1	2	3	4	5

column

attention_mask (B, T)

1	1	1	1	1	1
0	0	1	1	1	1
0	0	0	1	1	1
0	1	1	1	1	1
0	1	1	1	1	1
0	1	1	1	1	1

seq 1: score mask (T, T)

	0	0	1	1	1	1
[PAD]	-inf	-inf	-inf	-inf	-inf	-inf
[PAD]	-inf	-inf	-inf	-inf	-inf	-inf
I	-inf	-inf	0.0	-inf	-inf	-inf
love	-inf	-inf	0.0	0.0	-inf	-inf
deep	-inf	-inf	0.0	0.0	0.0	-inf
learning	-inf	-inf	0.0	0.0	0.0	0.0
	[PAD]	[PAD]	I	love	deep	learning

query

key

Demo time!!

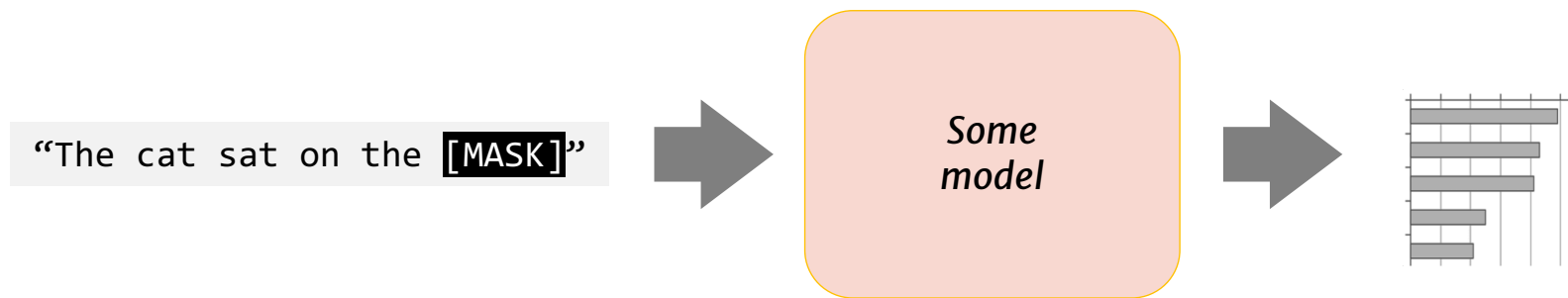
- <https://poloclub.github.io/transformer-explainer/>



Transformer Architectural Variants

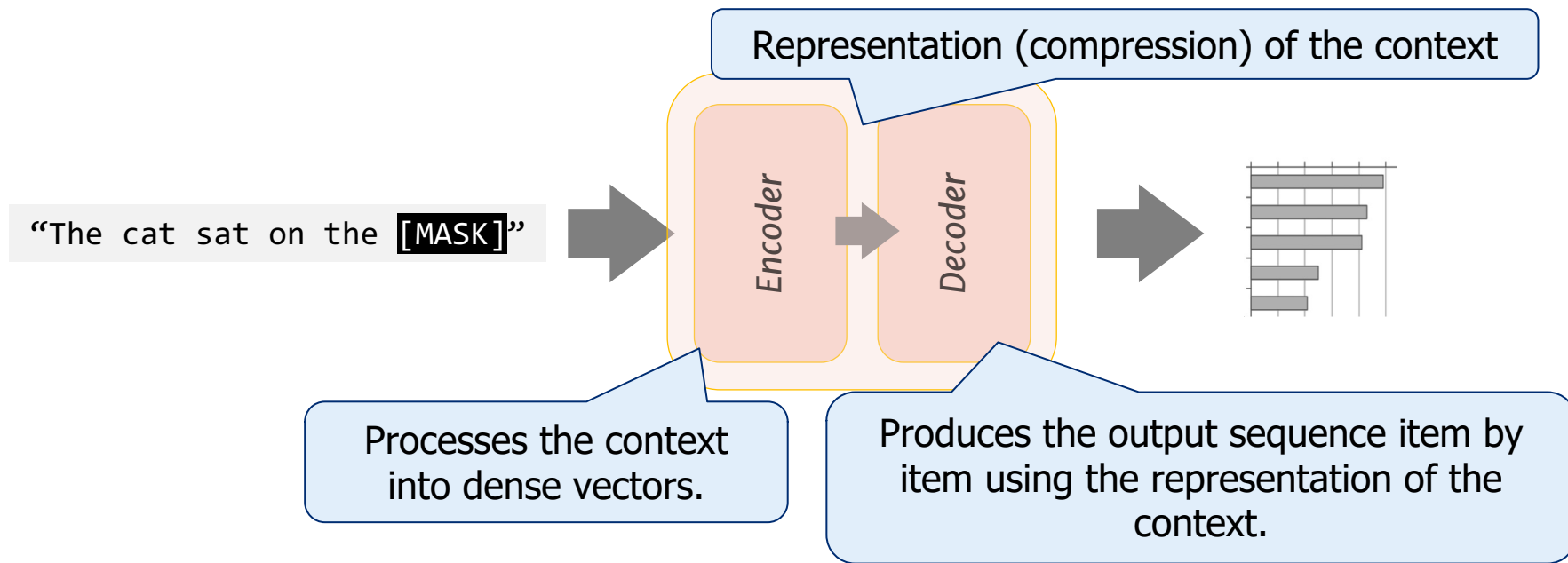
Encoder-Decoder Architectures

- It is useful to think of generative models as two sub-models.



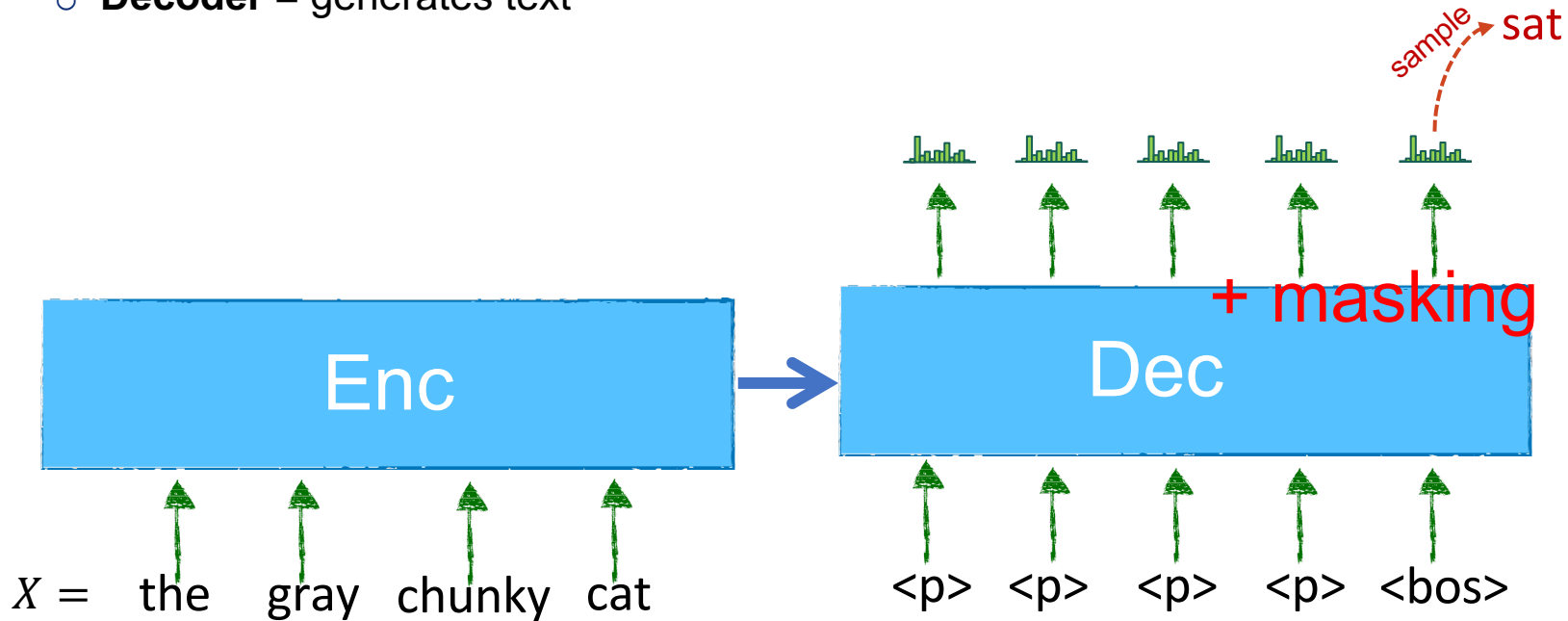
Encoder-Decoder Architectures

- It is useful to think of generative models as two sub-models.



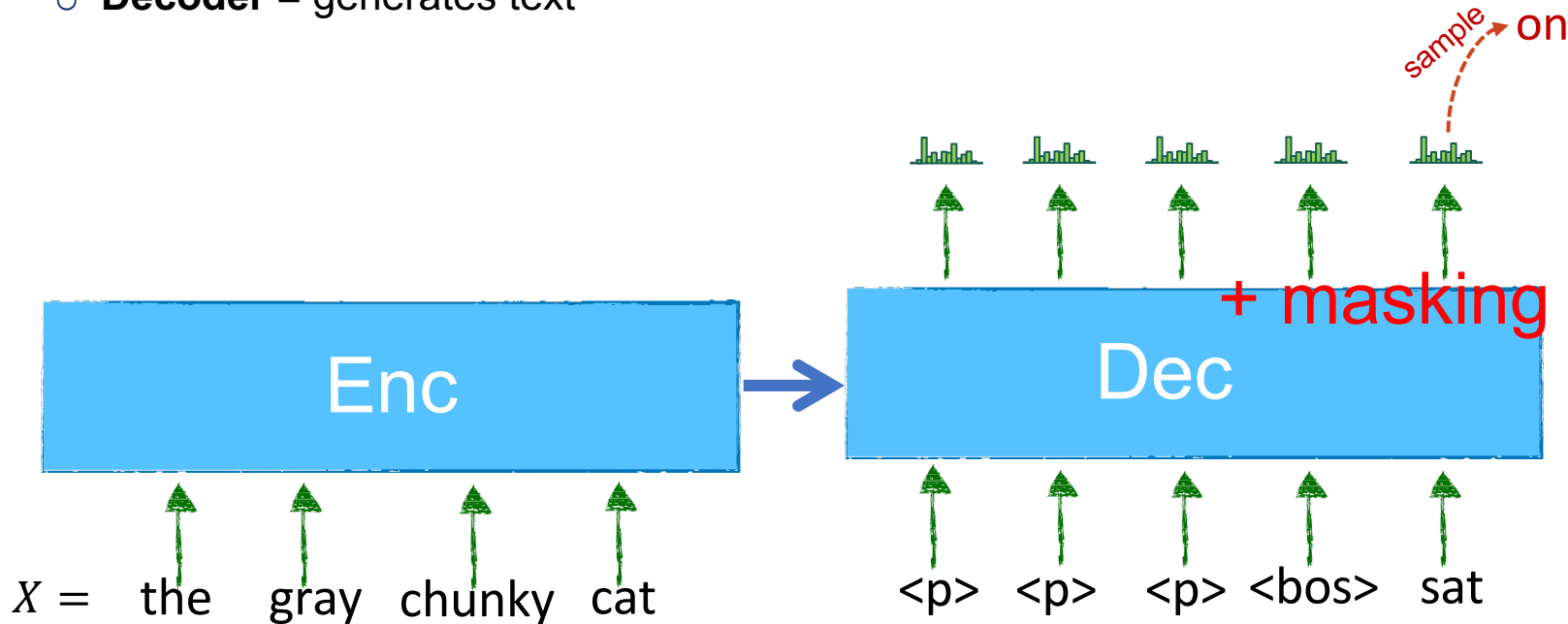
Enc-Dec work at inference time

- Transformer is two blocks
 - Encoder** = read or encode the input
 - Decoder** = generates text



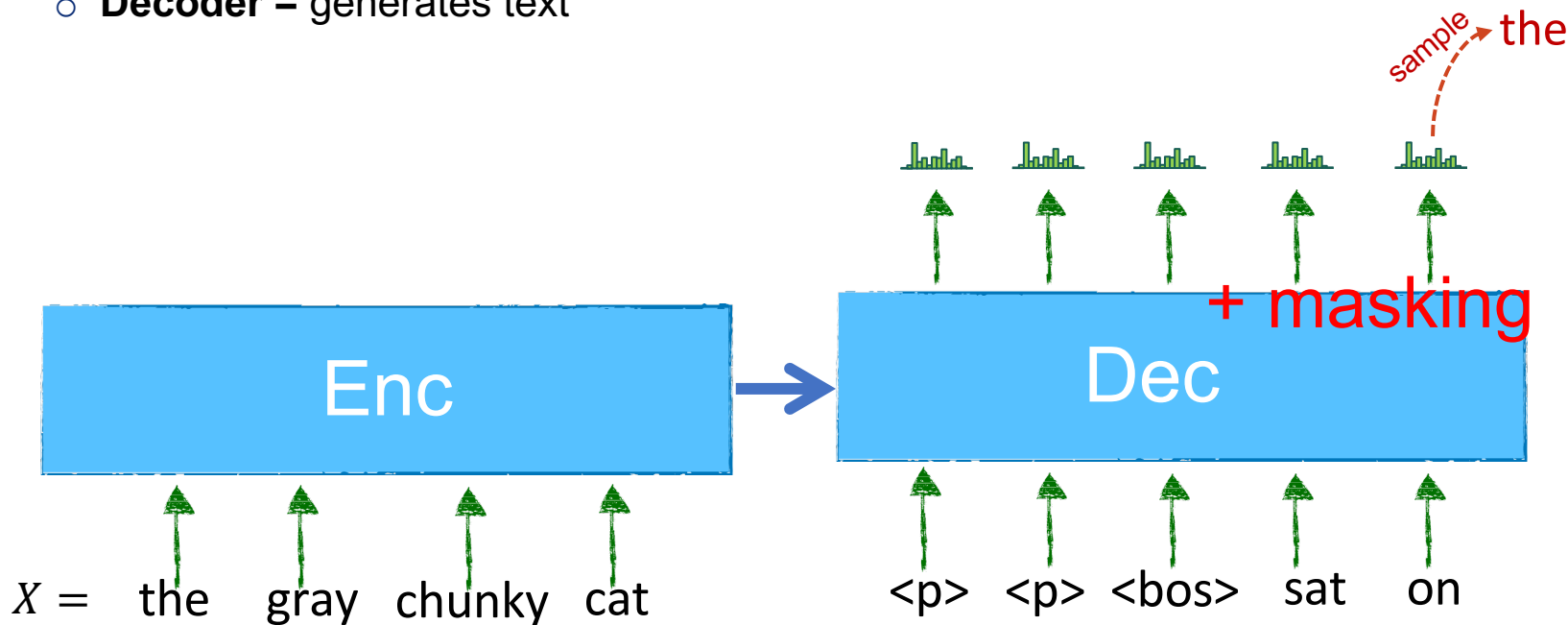
Enc-Dec work at inference time

- Transformer is two blocks
 - Encoder** = read or encode the input
 - Decoder** = generates text



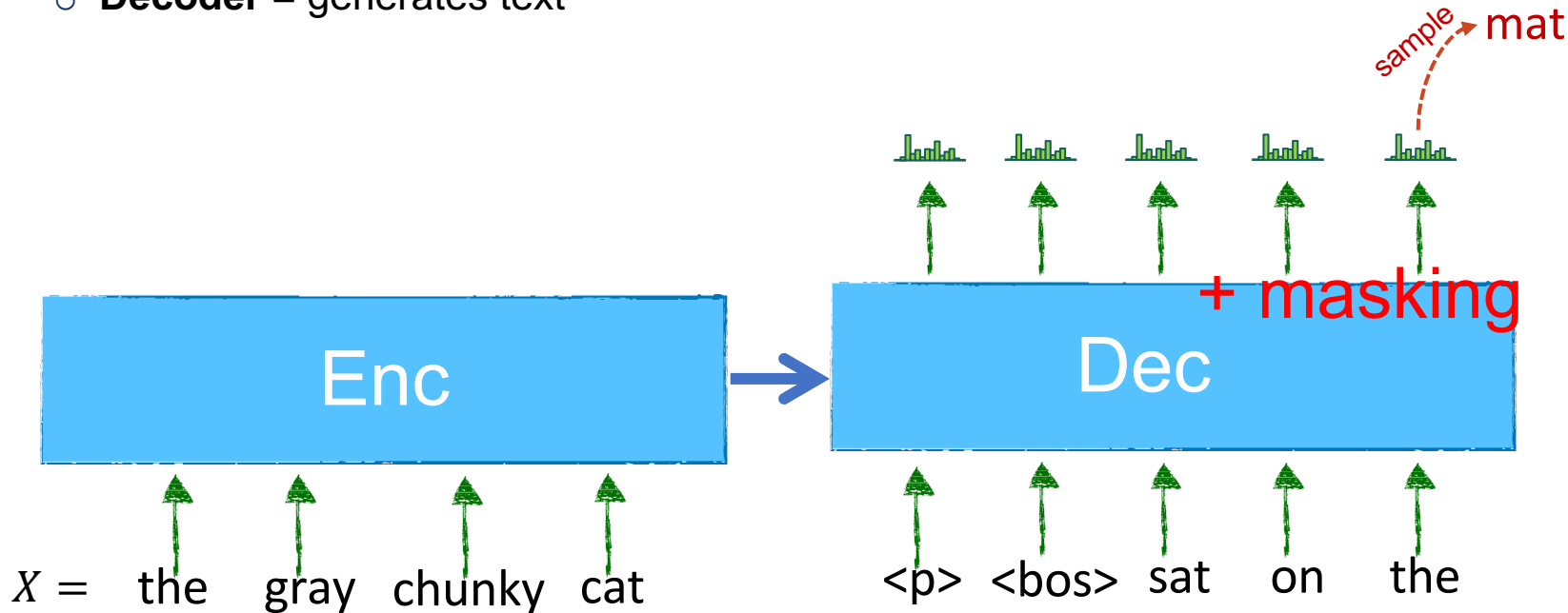
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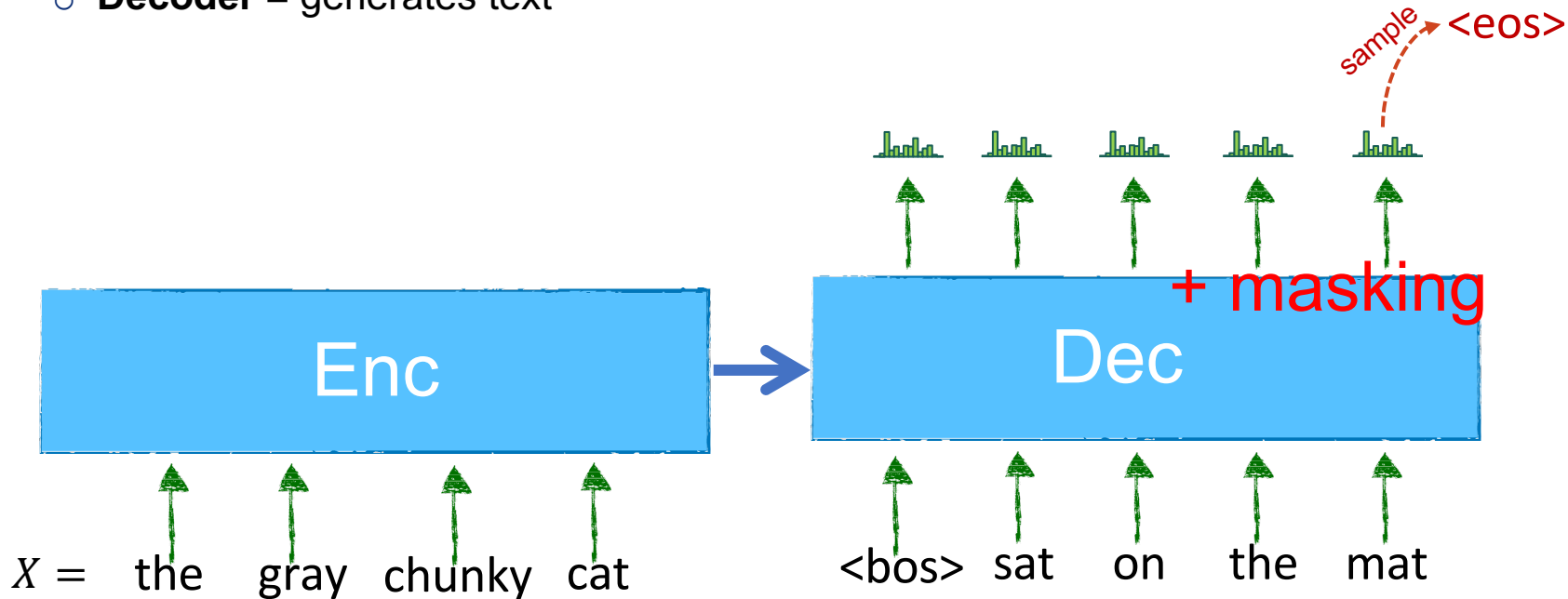
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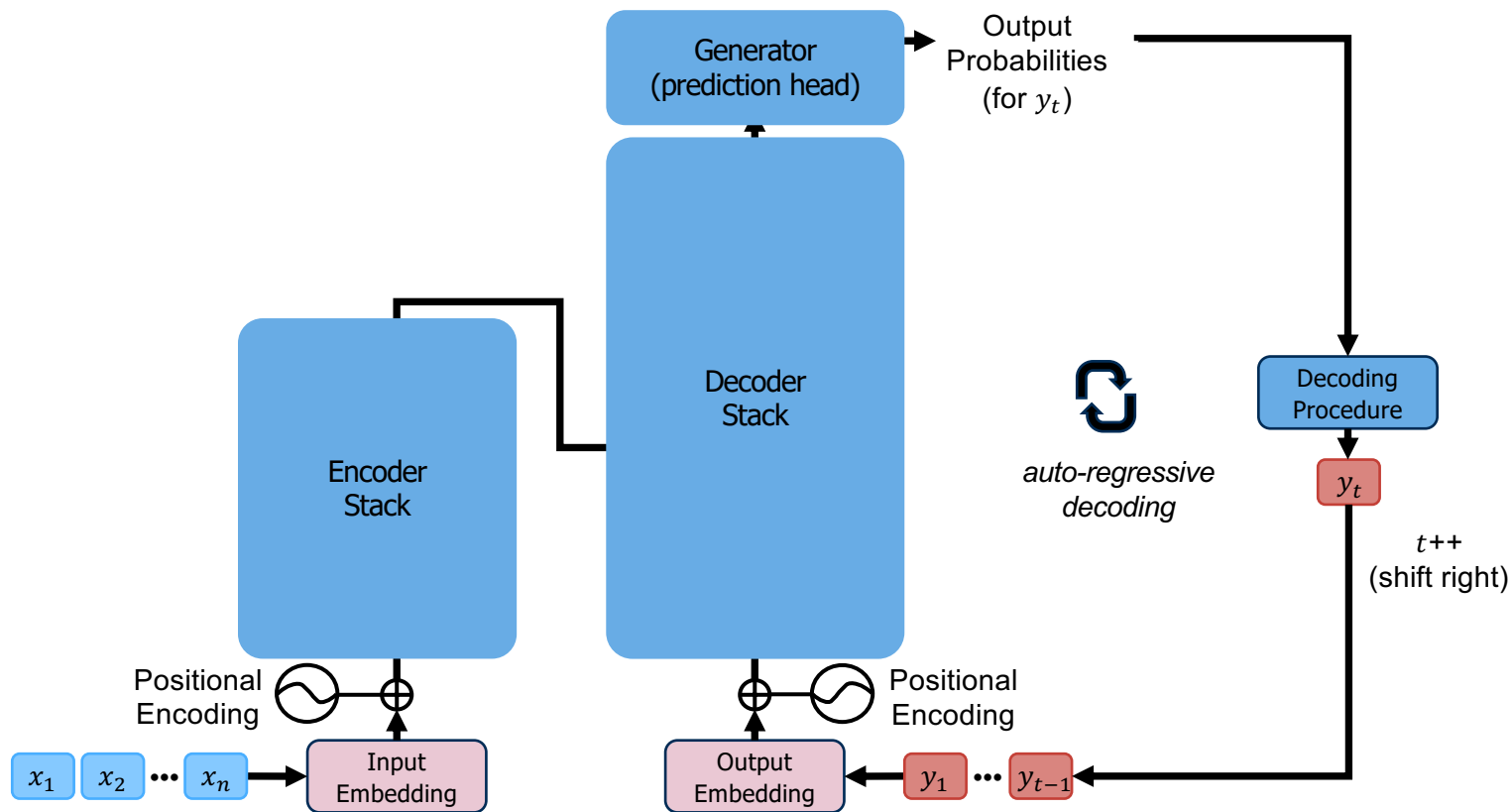


Enc-Dec work at inference time

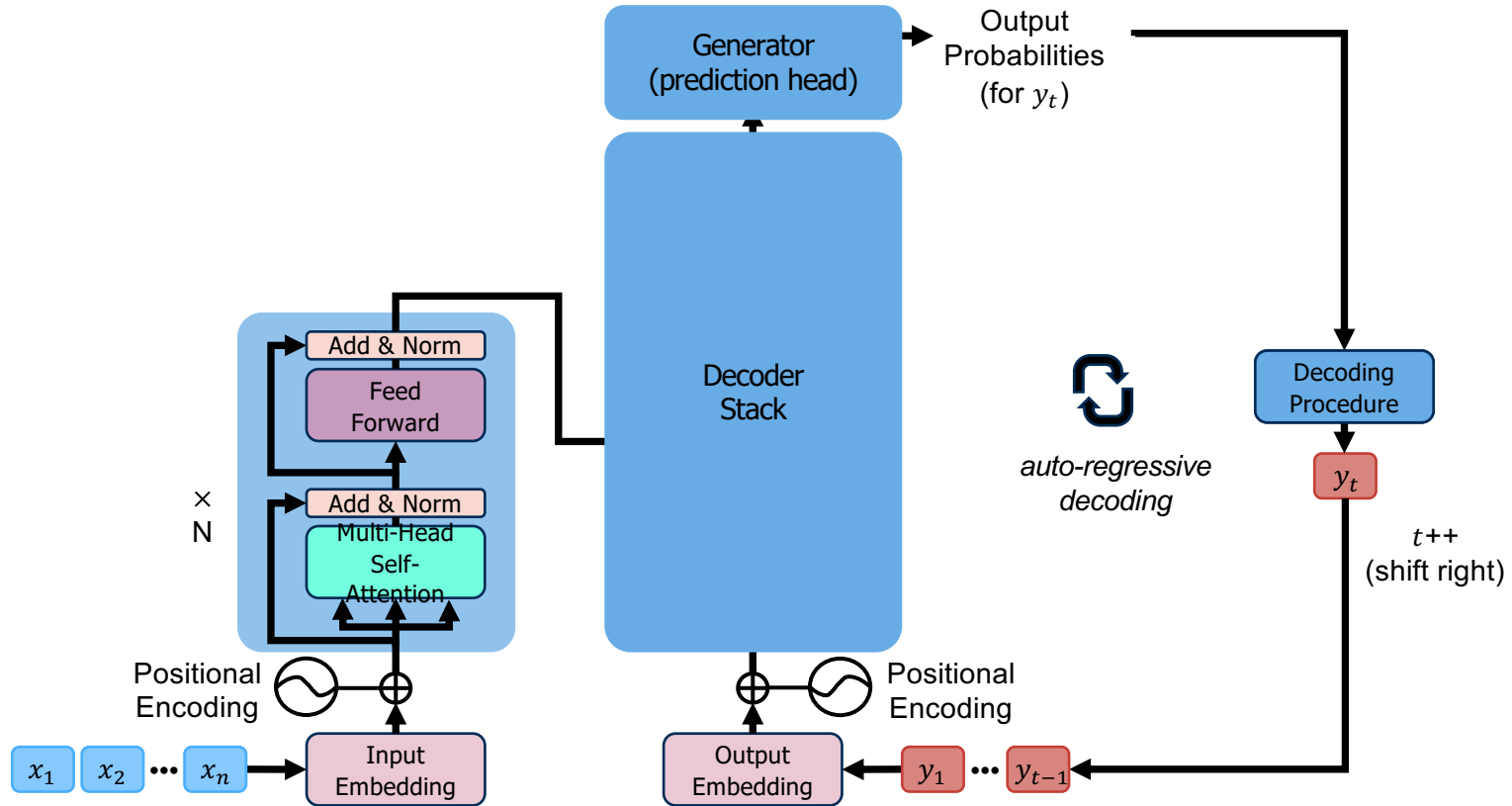
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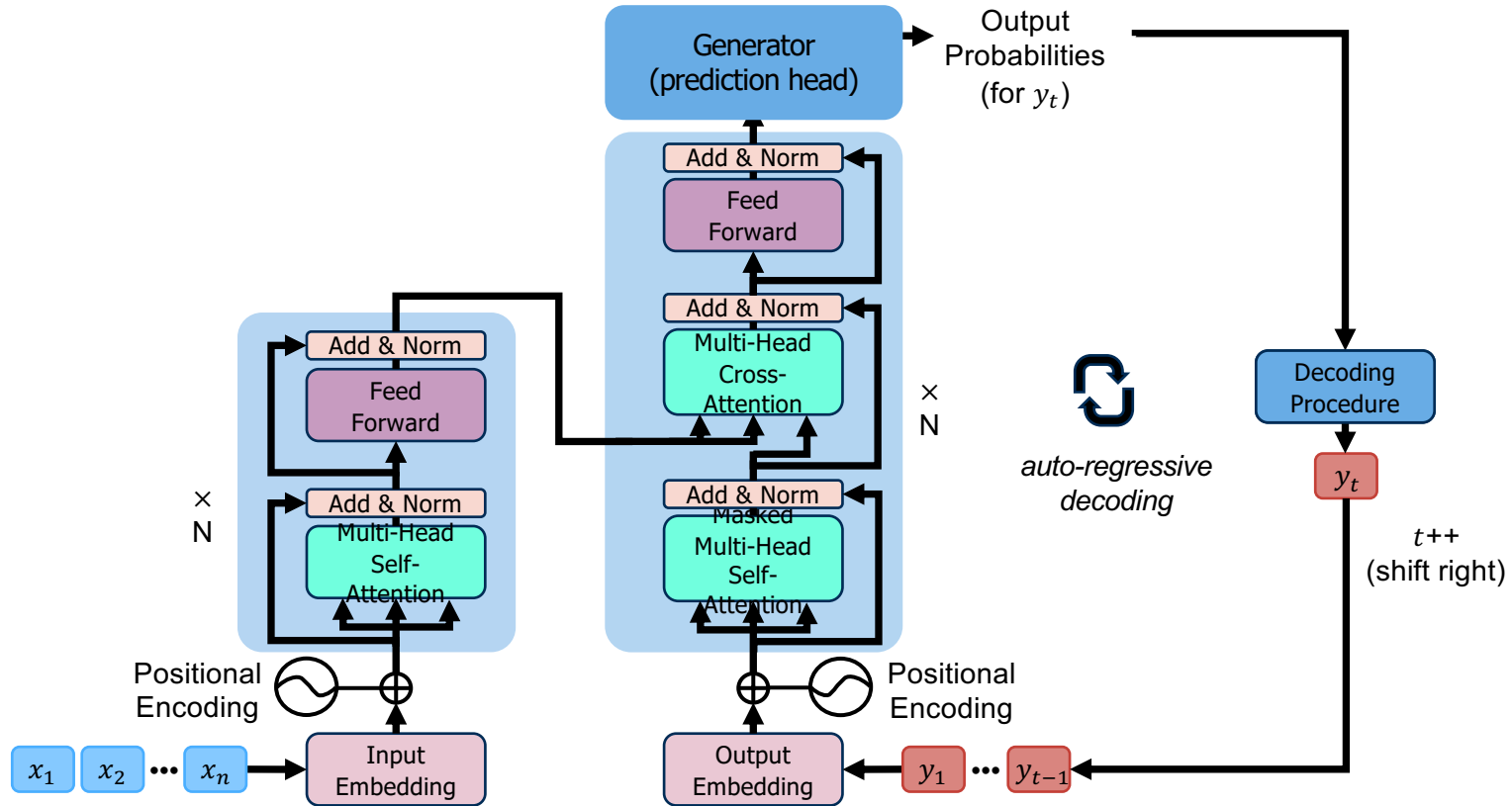
Transformer [Vaswani et al. 2017]



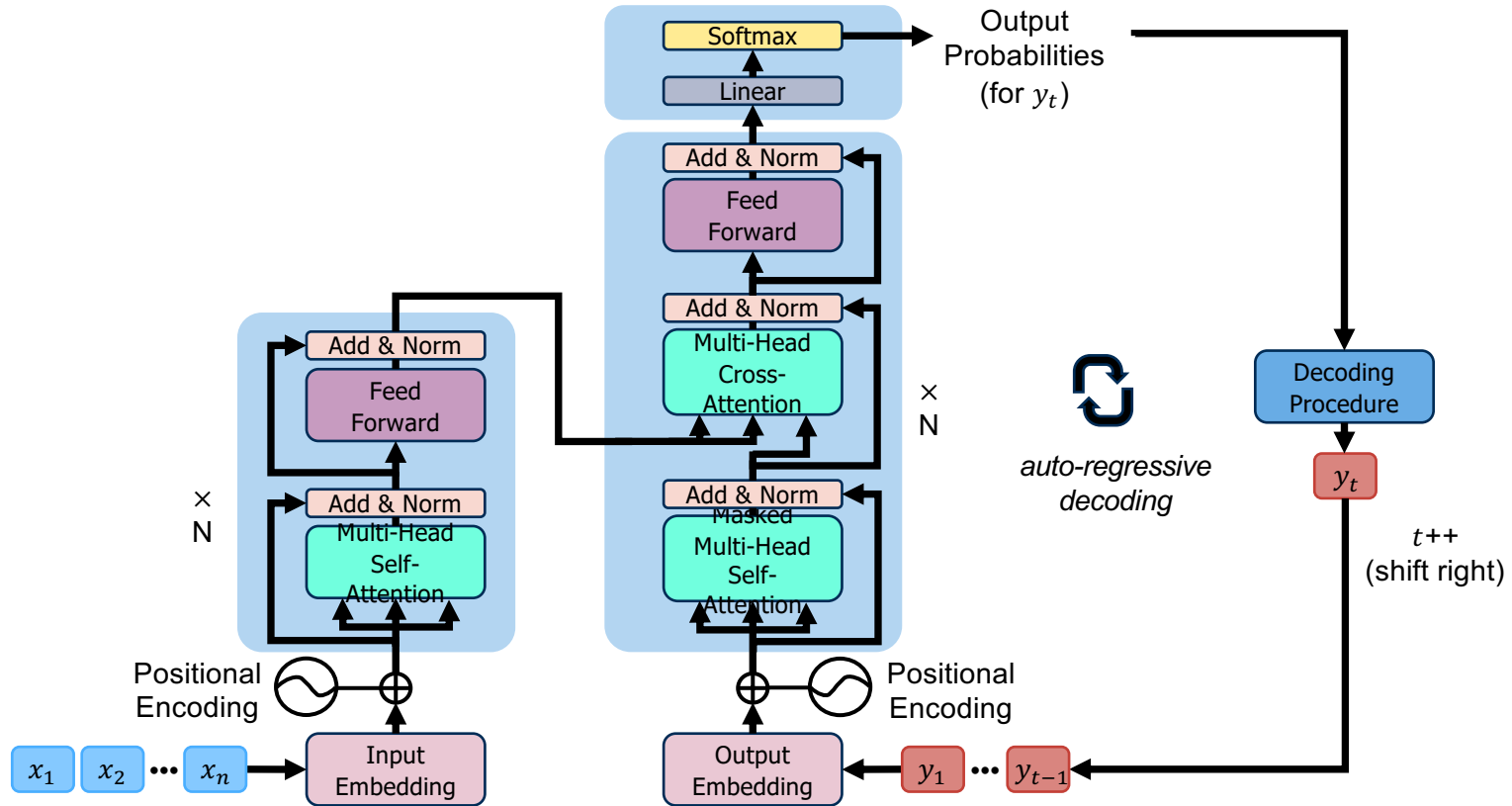
Transformer [Vaswani et al. 2017]



Transformer [Vaswani et al. 2017]

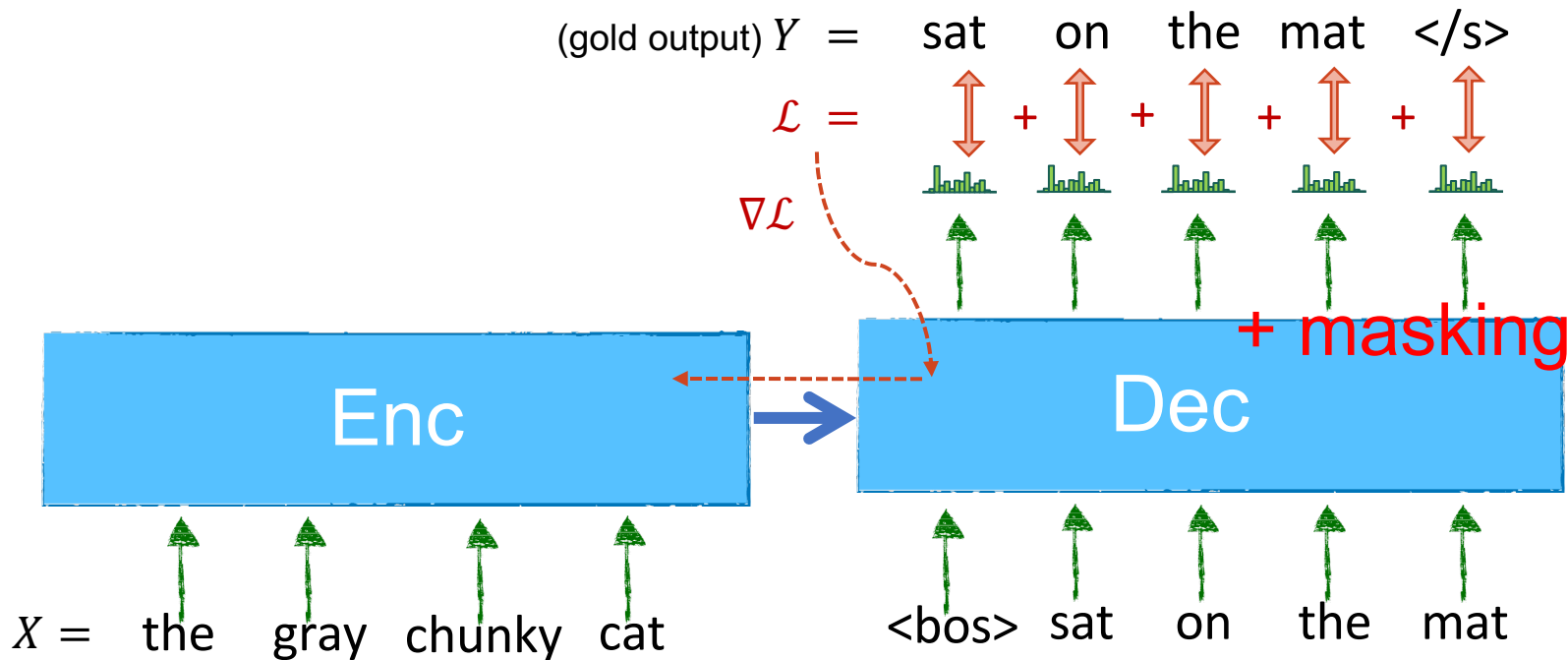


Transformer [Vaswani et al. 2017]



How does training Enc-Dec work?

- Same as what we did before for encoder-only models.

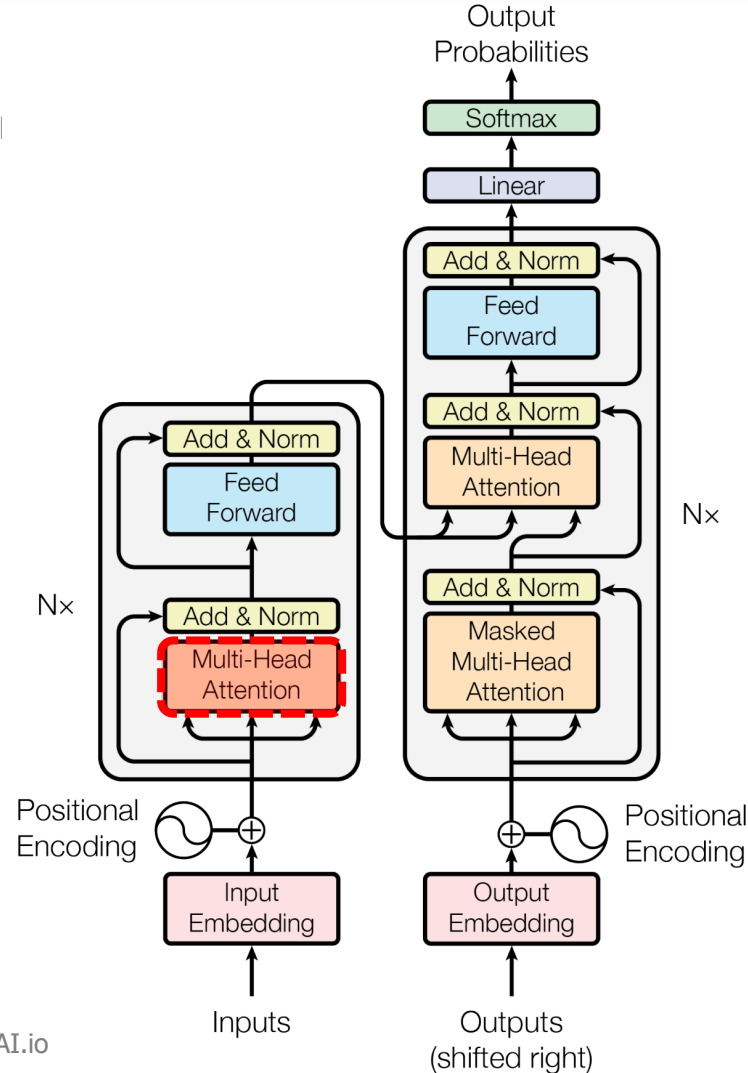
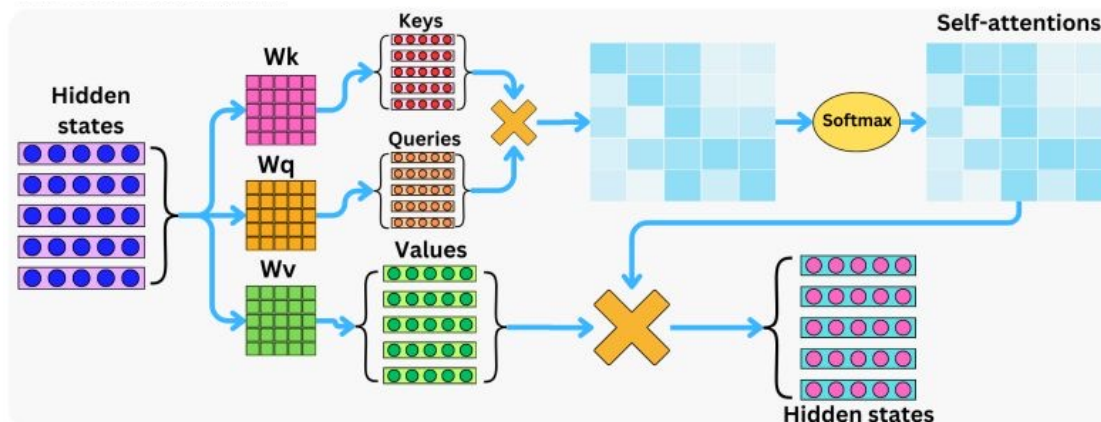


Transformer [Vaswani

- Computation of **encoder** attends to both sides.

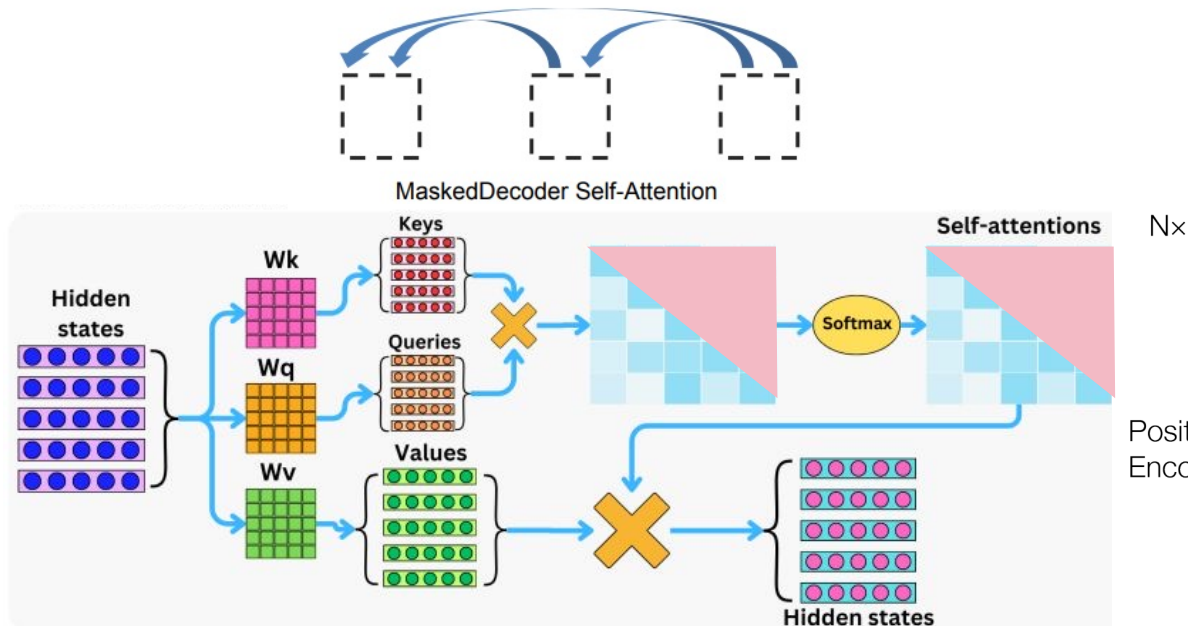


Encoder Self-Attention



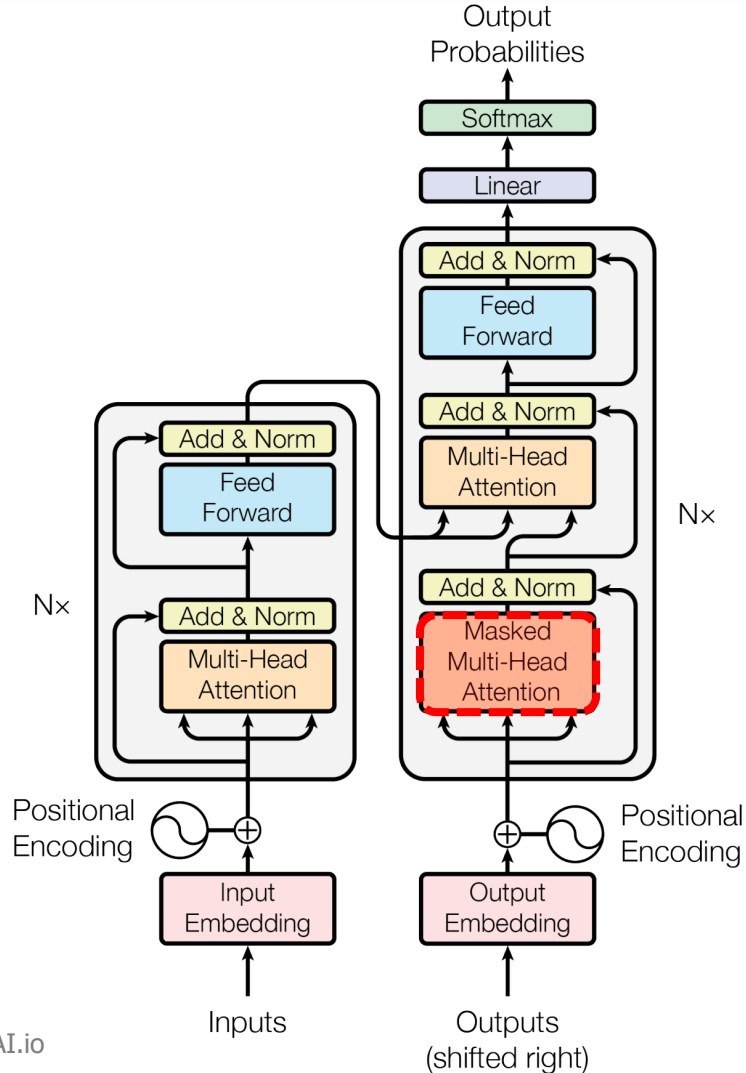
Transformer [Vaswani

- At any step of **decoder**, it attends to previous computation of **encoder** as well as **decoder's** own generations.



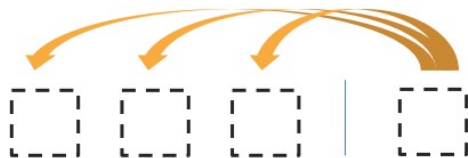
[[Attention Is All You Need, Vaswani et al. 2017](#)]

Visualization: theEdgeAI.io

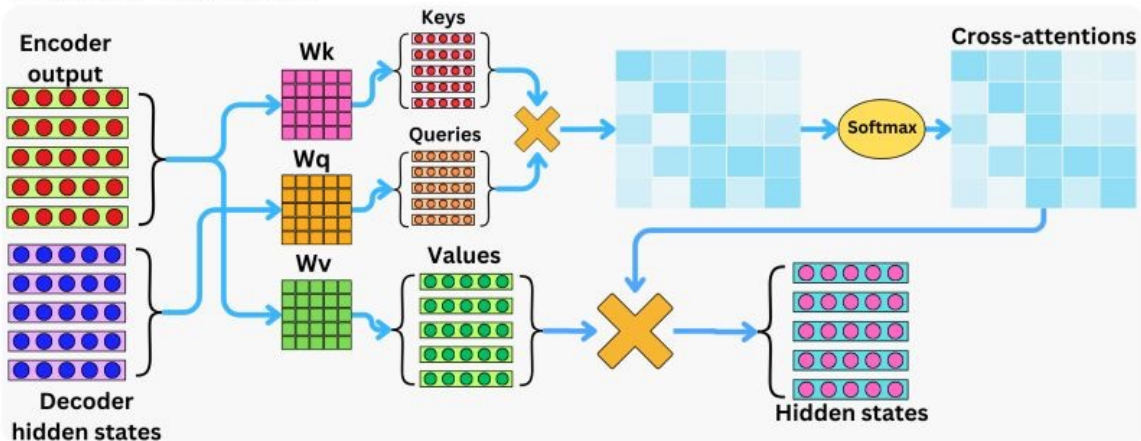


Transformer [Vaswani

- At any step of **decoder**, it attends to previous computation of **encoder**.

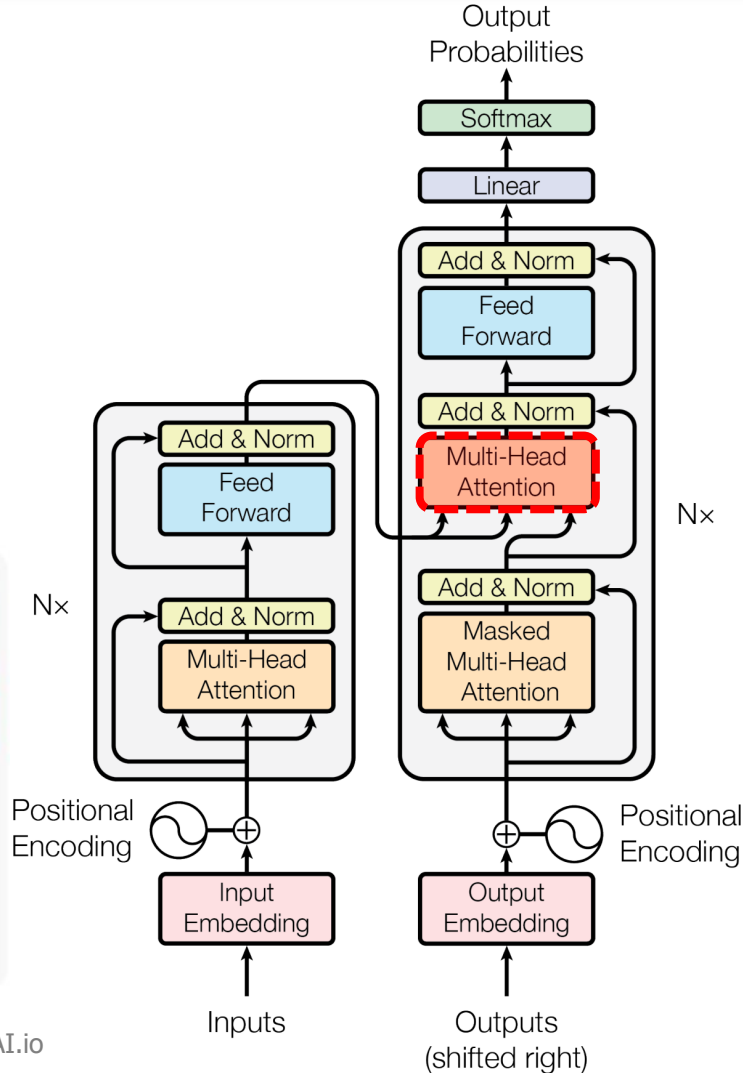


Encoder-Decoder Attention



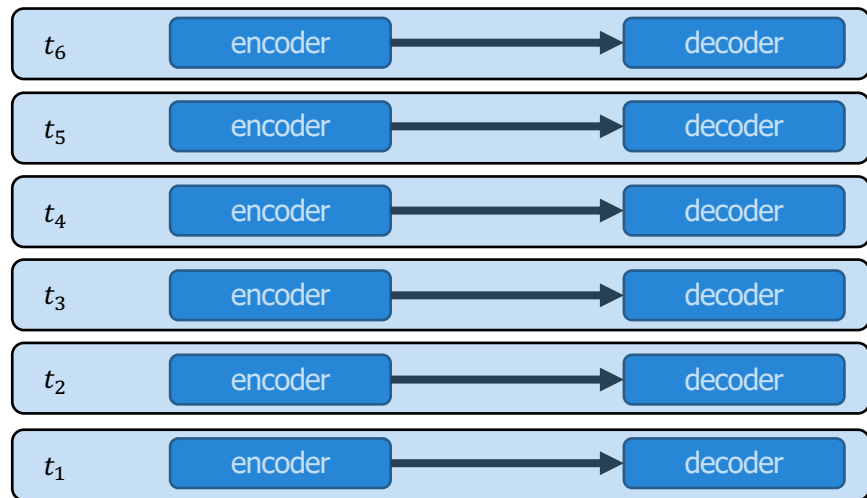
[[Attention Is All You Need, Vaswani et al. 2017](#)]

Visualization: theEdgeAI.io

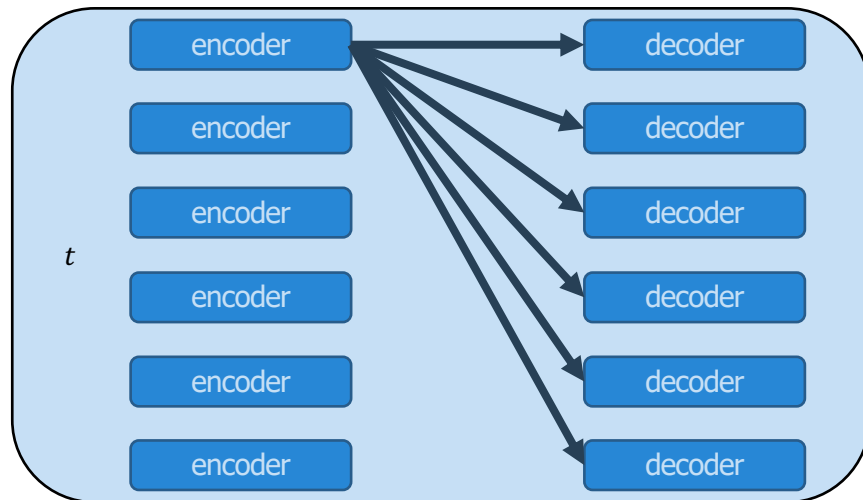


Quiz: Enc-Dec Connections

- Which best represents encoder-decoder connections? (y-axis denotes the layers)



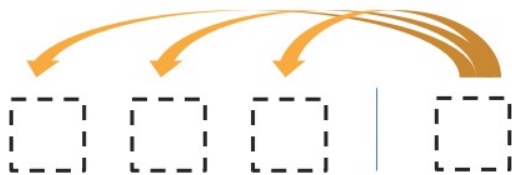
Incorrect



Correct

Recap: Transformer

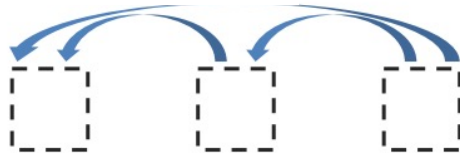
- Yaaay we know Transformers now! 🥳
- An **encoder-decoder** architecture
- 3 forms of attention



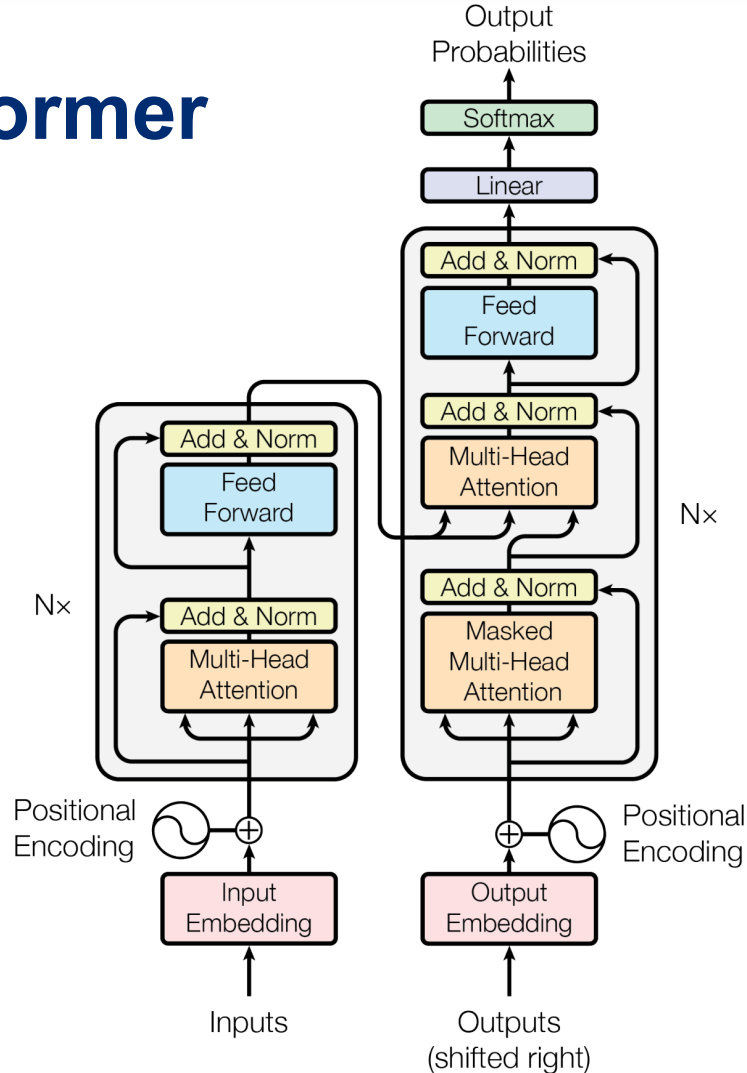
Encoder-Decoder Attention



Encoder Self-Attention



Masked Decoder Self-Attention



Considerations about computational cost in Transformers

Diversion: Floating-point Ops: FLOPS

- Floating point operations per second (FLOPS, flops or flop/s)
- Each FLOP can represent an addition, subtraction, multiplication, or division of floating-point numbers,
- The total FLOP of a model (e.g., Transformer) provides a basic approximation of computational costs associated with that model.

FLOPS of Matrix Multiplication

- Matrix-vector multiplication are common in Self-Attention (e.g., QKV projection)
 - Requires $2mn$ (2 x matrix size) operations for multiplying $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^n$
 - (2 because 1 for multiplication, 1 for addition)

$$\begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \cdots & A_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} A_{11}x_1 + A_{12}x_2 + \cdots + A_{1n}x_n \\ A_{21}x_1 + A_{22}x_2 + \cdots + A_{2n}x_n \\ \vdots \\ A_{m1}x_1 + A_{m2}x_2 + \cdots + A_{mn}x_n \end{bmatrix}$$

Quiz: thinking about computations

- Consider the following matrix multiplication:

$$A B, \quad \text{where } A \in \mathbb{R}^{n \times m}, B \in \mathbb{R}^{m \times k}$$

- **Question 1:** Computing AB involves how many **arithmetic operations**?
 - (also referred to as floating-point operations or FLOPs)
- **Answer:** It's $O(n \times m \times k)$.
 - (to be a bit more precise, it's $\approx 2n \times m \times k$ since each element in AB requires almost equal num of multiplications and summations.)
- **Question 2:** Computing AB involves how many memory/IO access?
- **Answer:** It's $n \times m$ (reading A) + $m \times k$ (reading B) + $n \times k$ (writing AB).

(*assuming each input is read once)

Computations in Transformer Block

- We are going to count computations and IO access in Transformer computations.
- Note we assume that the full input sequence is given at once (e.g., training time).
- Here is the first step. Given: $\mathbf{x} \in \mathbb{R}^{b \times n \times d}$, $\mathbf{W}_i^q \in \mathbb{R}^{d \times \frac{d}{m}}$
- Let's think about the following computation: $\mathbf{x}\mathbf{W}_i^q$
- **Q1:** What is the number of arithmetic operations? $O(b \times n \times d \times \frac{d}{m})$ for each head
- **Q2:** What about the number of IO? $O(b \times n \times d + d \times \frac{d}{m} + b \times n \times \frac{d}{m})$ for each head
- **Q3:** What are these quantities for all heads?
 - Number of arithmetic ops: $O(bnd^2)$
 - Number of IO ops: $O(2bnd + d^2)$

Computations in Transformer Block

b : batch size,
 n : sequence length,
 m : number of heads
 d : feature dimension in output of SA
 d/m : feature dimension inside each SA head
 $d_{\text{ff}} = 4d$: feature dimension inside FFN

Dimensions	Operation	Computations	IO
$\mathbf{x} \in \mathbb{R}^{b \times n \times d}, \mathbf{W}_i^q \in \mathbb{R}^{d \times \frac{d}{m}}$	$\mathbf{x}\mathbf{W}_i^q, \mathbf{x}\mathbf{W}_i^k, \mathbf{x}\mathbf{W}_i^v$ for m heads	$O(bnd^2)$	$O(d^2 + 2bnd)$
$\mathbf{Q}_i, \mathbf{K}_i \in \mathbb{R}^{b \times n \times \frac{d}{m}}$	$P_i \leftarrow \text{softmax}\left(\frac{\mathbf{Q}_i \mathbf{K}_i^T}{\sqrt{d/m}}\right)$ for m heads	$O(bn^2d)$	$O(2bnd + bmn^2)$
$\mathbf{V}_i \in \mathbb{R}^{b \times n \times \frac{d}{m}}, P_i \in \mathbb{R}^{b \times n \times n}$	$\text{head}_i \leftarrow P_i \mathbf{V}_i$ for m heads	$O(bn^2d)$	$O(2bnd + bmn^2)$
$\mathbf{W}^o \in \mathbb{R}^{d \times d}, \text{head}_i \in \mathbb{R}^{b \times n \times \frac{d}{m}}$	$Y = \text{Concat}(\text{head}_1, \dots, \text{head}_m) \mathbf{W}^o$	$O(bnd^2)$	$O(2bnd + d^2)$
$Y \in \mathbb{R}^{b \times n \times d}, \mathbf{W}_1 \in \mathbb{R}^{d \times d_{\text{ff}}}, \mathbf{W}_2 \in \mathbb{R}^{d_{\text{ff}} \times d}$	$Y = f(Y\mathbf{W}_1)\mathbf{W}_2$	$O(16bnd^2)$	$O(2bnd + 8d^2)$
---	Total	$O(bnd^2 + bn^2d)$	$O(bnd + bmn^2 + d^2)$

The bottlenecks

b : batch size,
 n : sequence length,
 d : feature dimension in output of SA

- So, in total, we have $\rightarrow$
- The quadratic terms are based on n and d
- d is fixed (part of architecture) but n changes with input.
- **Bottlenecks #1:** If n (sequence length) $\gg d$ (feature dimension), the time and space complexity would be dominated by $O(n^2)$.
- **However**, despite this quadratic dependence these are parallelizable operations which can be computed efficiently in GPUs.
 - In comparison, RNNs perform less arithmetic ops but they're not all parallelizable.
- **Bottlenecks #2:** Another potential bottleneck is how fast we can run IO.
(more on this later)

Computations	IO
$O(bnd^2 + bn^2d)$	$O(bnd + bmn^2 + d^2)$

Layer Type	Complexity per Layer	Sequential Operations
Self-Attention	$O(n^2 \cdot d)$	$O(1)$
Recurrent	$O(n \cdot d^2)$	$O(n)$

[[Vaswani et al. 2017](#)]

Recap

- Transformers **computation of a full sequence** is bounded by $O(bnd^2 + bn^2d)$.
 - Generally, the quadratic term that depends on seq len n is more concerning.
- **IO may also impose** other limits on this (coming up).
- Also, the above calculations is for a given sentences (e.g. training time).
 - How bad is the computational complexity during the decoding time where we want to generate text one token at a time?



During decoding time, how slow is
attention computation?



Q: Pulse Check: What is a KV-Cache?

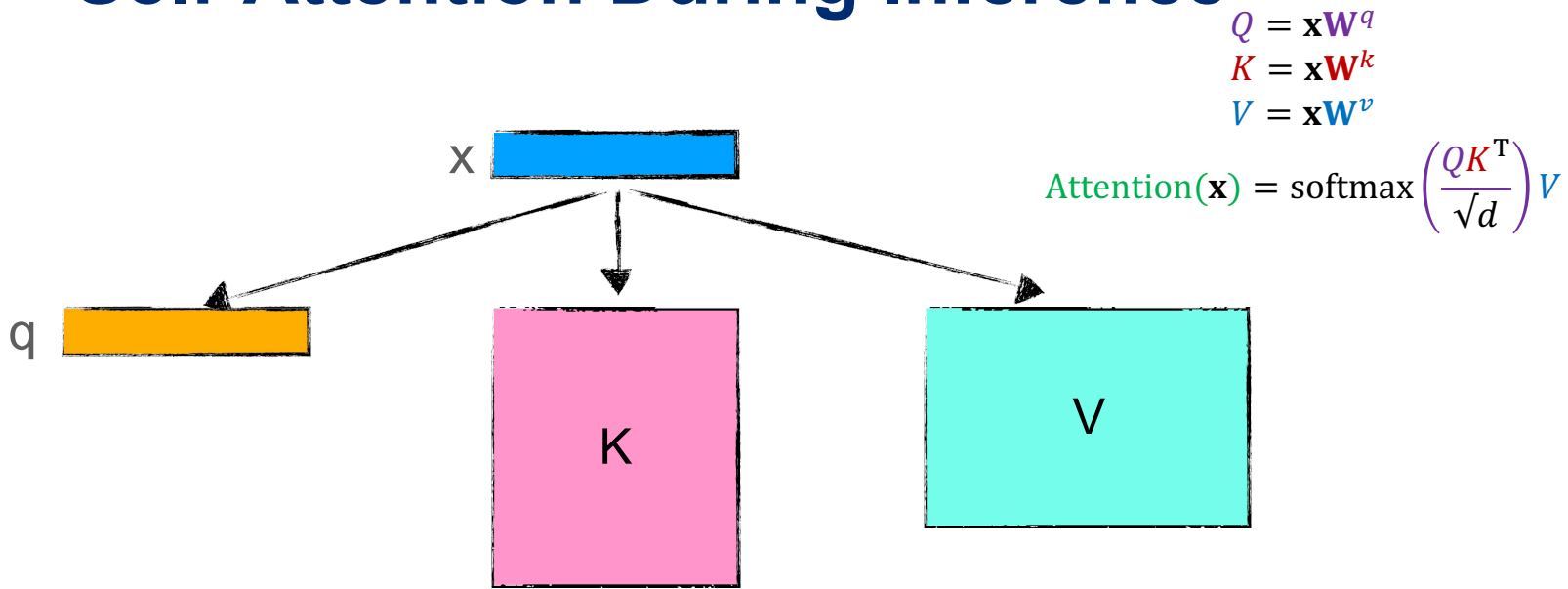
$$Q = \mathbf{x} \mathbf{W}^q$$

$$K = \mathbf{x} \mathbf{W}^k$$

$$V = \mathbf{x} \mathbf{W}^v$$

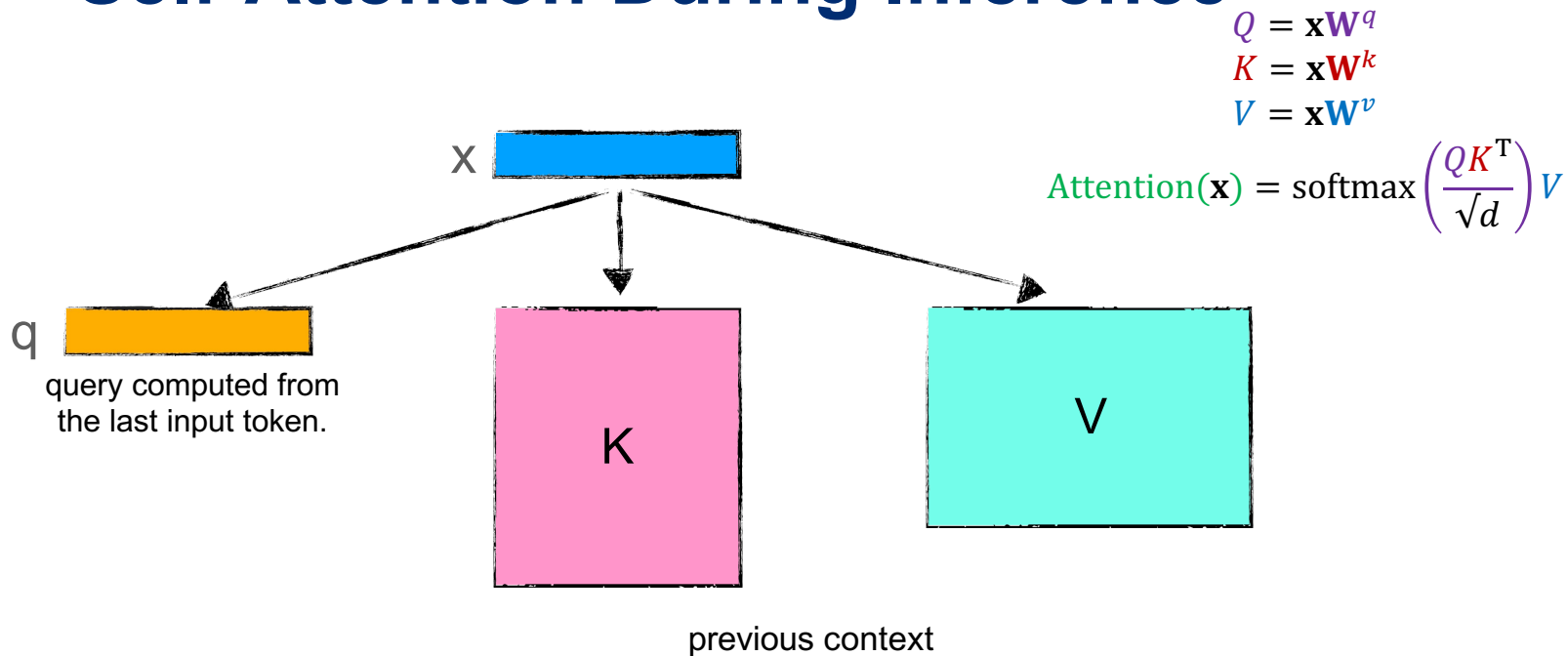
$$\text{Attention}(\mathbf{x}) = \text{softmax} \left(\frac{QK^T}{\sqrt{d}} \right) V$$

Self-Attention During Inference



[Slide credit: Arman Cohan]

Self-Attention During Inference



[Slide credit: Arman Cohan]

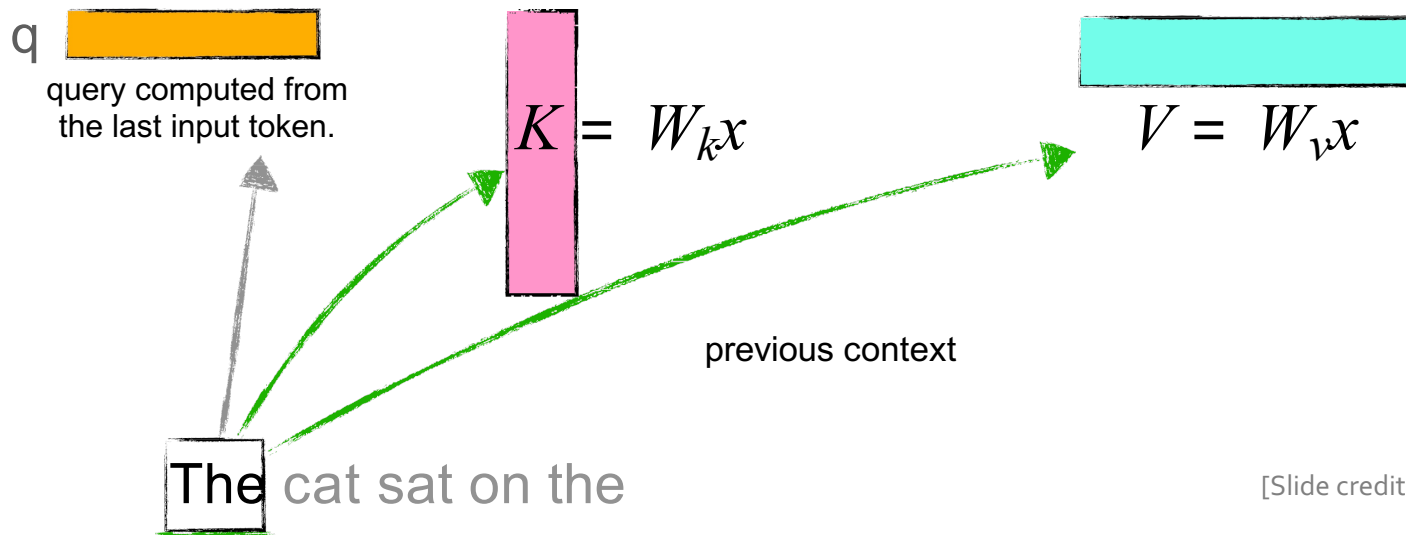
Self-Attention During Inference

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$$\text{Attention}(\mathbf{x}) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V$$



[Slide credit: Arman Cohan]

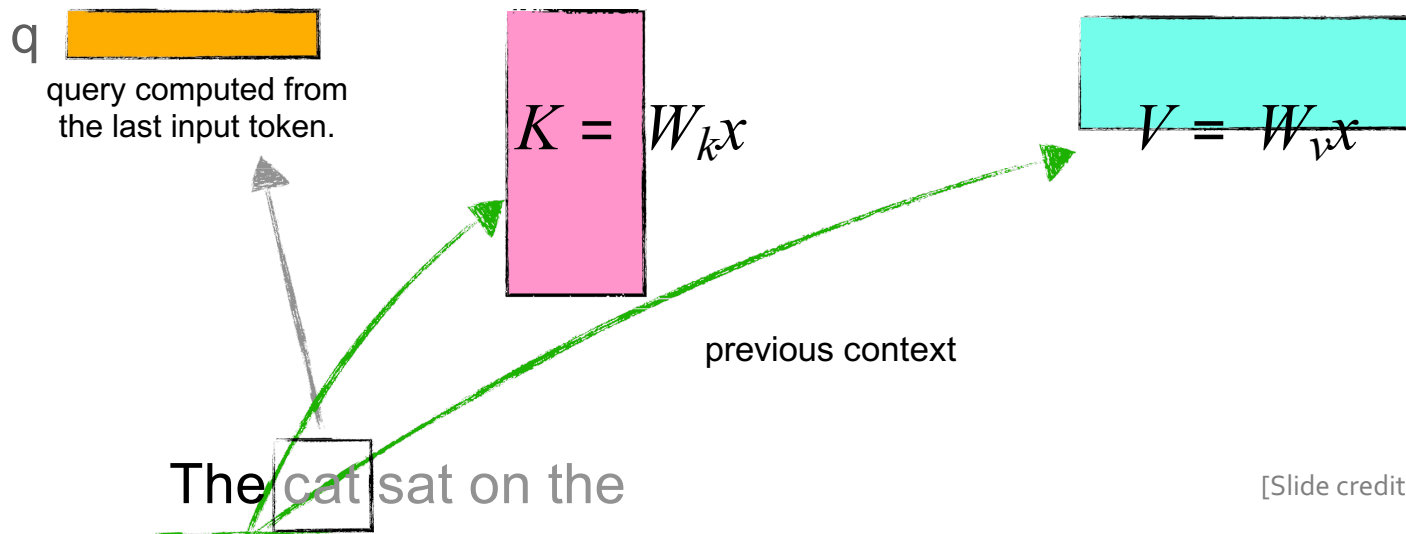
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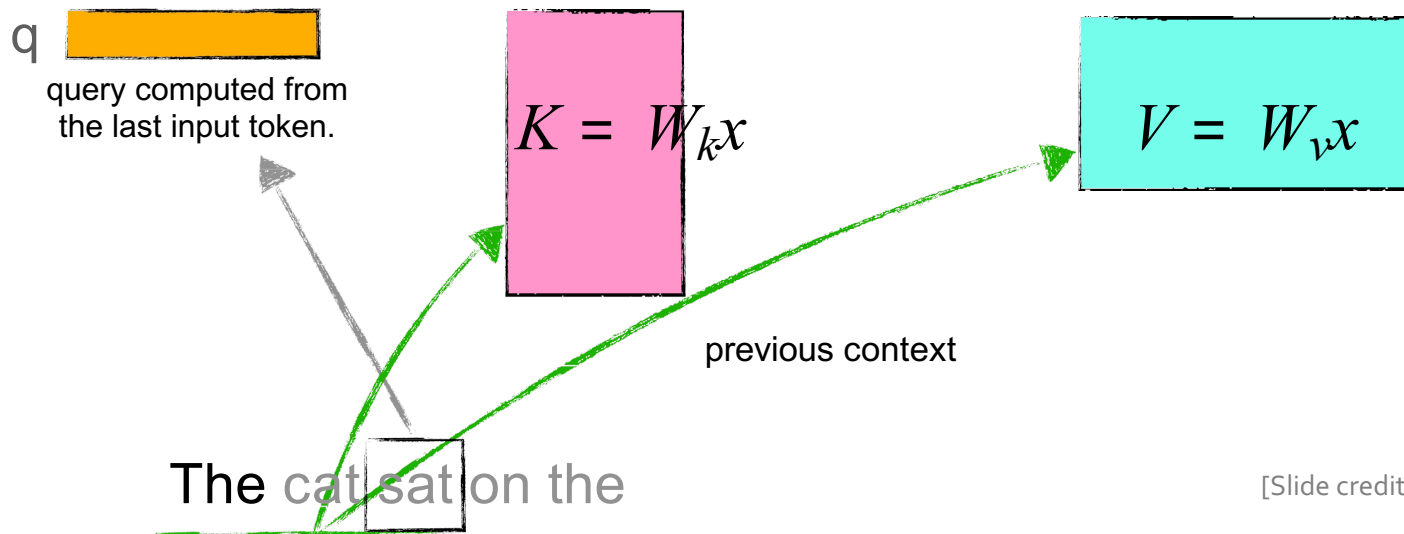
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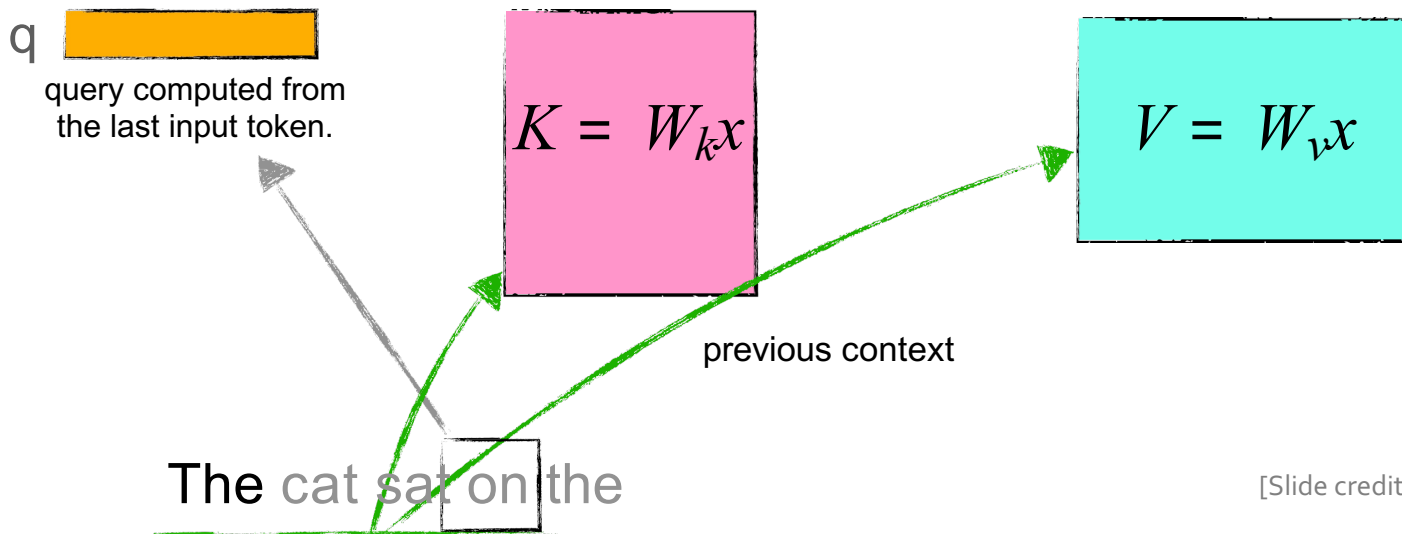
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[Slide credit: Arman Cohan]

Self-Attention During Inference

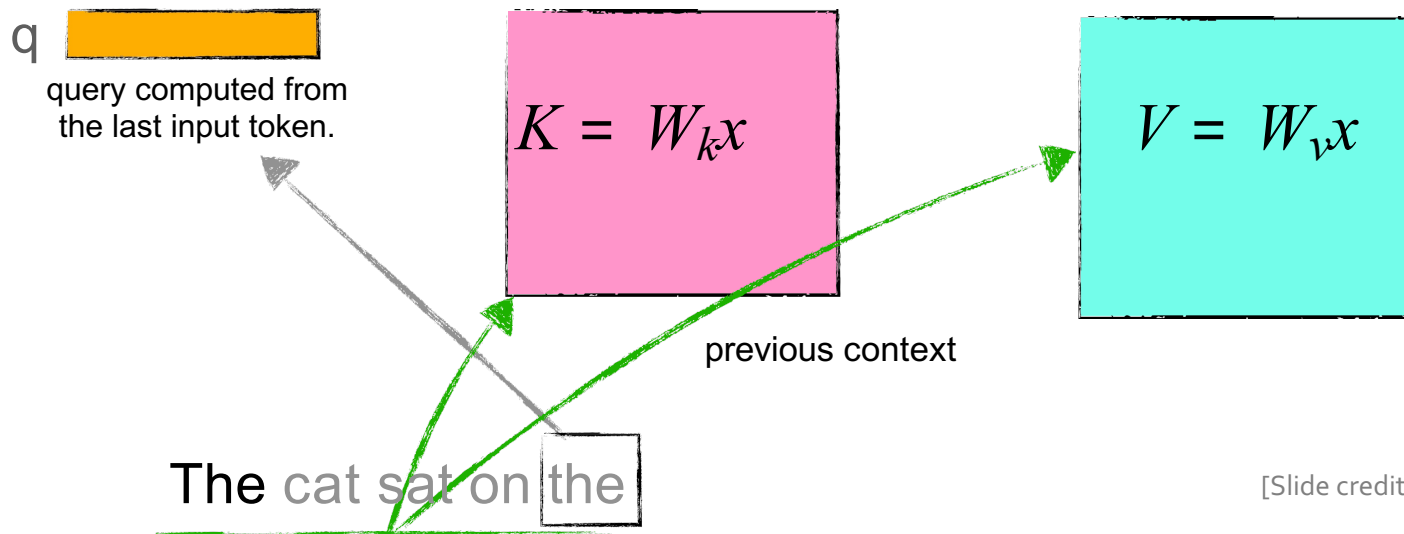
- We are computing the Keys and Values many times!
 - Let's reduce redundancy! 🤔

$$Q = \mathbf{x}W^q$$

$$K = \mathbf{x}W^k$$

$$V = \mathbf{x}W^v$$

$$\text{Attention}(\mathbf{x}) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V$$



[Slide credit: Arman Cohan]

KV-Cache for reducing inference redundancy

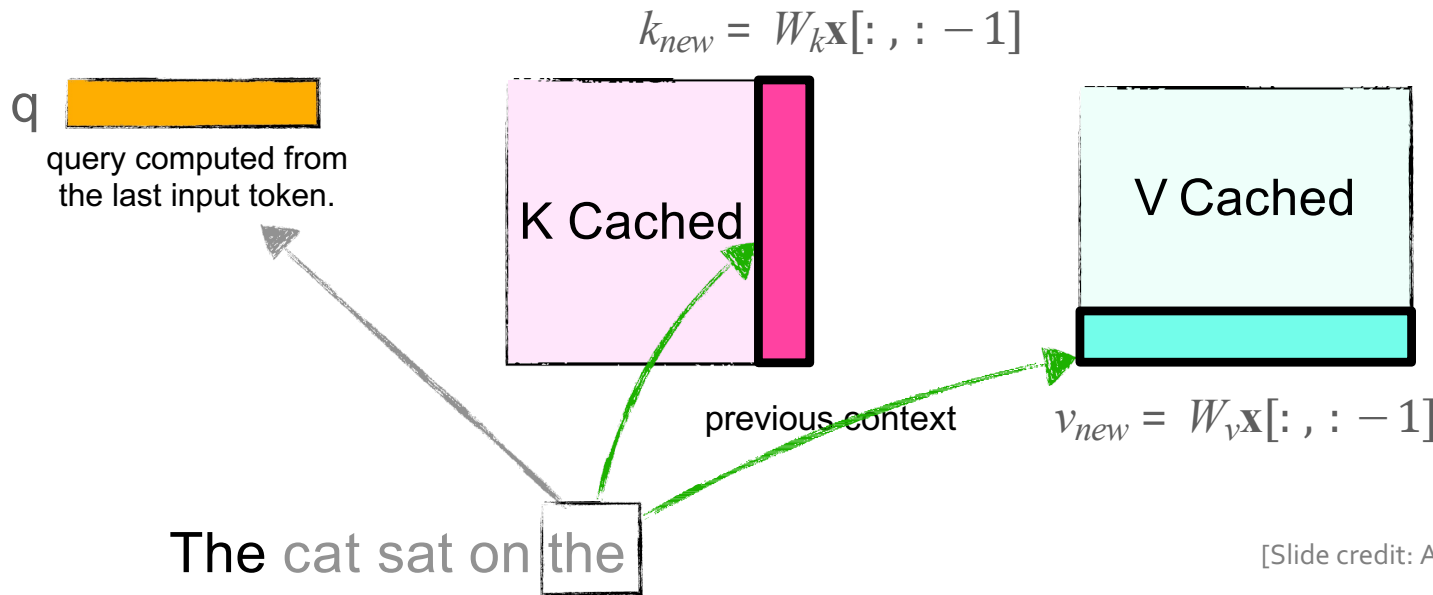
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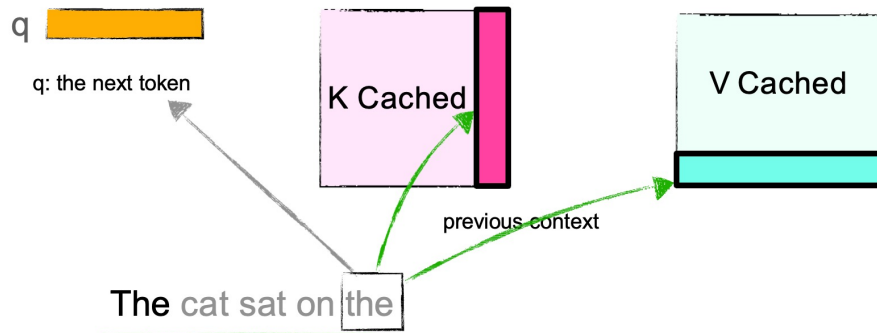


[Slide credit: Arman Cohan]

Quiz: KV-Cache

- How much **memory** does this KV-cache require? Let's assume,
 - batch size b ,
 - embedding dimension is d ,
 - the length of the sequence seen so far is n ,
 - your model has L layers,
 - Each param is stored k bytes (e.g., FP16 takes 2 bytes)

1. $2bnLk$
2. $2bdLk$
3. $bndLk$
4. $2bndLk$



Recap

- To avoid **redundant computations** during decoding time, KV-cache is used to keep track of previous calculations of keys and values.
- But how exactly how costly are these computations?

Decoding Computations (processing token by token)

b : batch size,
 n : sequence length **thus far**,
 m : number of heads
 d : feature dimension in output of SA
 $d_{ff} = 4d$: feature dimension inside FFN

Notice we're doing this computations for one token

Compared to the previous table (SA for a seq of length n), all the cells have one less dependence on n (e.g., $n^2 \rightarrow n$ or $n \rightarrow 1$).

Dimensions	Operation	Computations	IO
$\mathbf{x} \in \mathbb{R}^{b \times 1 \times d}$, $\mathbf{W}_i^q \in \mathbb{R}^{d \times \frac{d}{m}}$	Query/key computations get combined with KV-cache	$O(bd^2)$	$O(d^2 + 2bd)$
$\mathbf{Q}_i, \mathbf{K}_i \in \mathbb{R}^{b \times 1 \times \frac{d}{m}}$ of the new token; $\mathbf{K}_{kv} \in \mathbb{R}^{b \times n \times \frac{d}{m}}$ from KV-cache	$P_i \leftarrow \text{softmax}\left(\frac{\mathbf{Q}_i [\mathbf{K}_{kv}; \mathbf{K}_i]^T}{\sqrt{d/m}}\right)$ for m heads	$O(bnd)$	$O(bnm + bnd + bd)$
$\mathbf{V}_i \in \mathbb{R}^{b \times 1 \times \frac{d}{m}}$ of the new token; $\mathbf{V}_{kv} \in \mathbb{R}^{b \times n \times \frac{d}{m}}$ from KV-cache; $P_i \in \mathbb{R}^{b \times 1 \times (n+1)}$	$\text{head}_i \leftarrow P_i [\mathbf{V}_{kv}; \mathbf{V}_i]$ for m heads	$O(bnd)$	$O(bnm + bnd + bd)$
$\mathbf{W}^o \in \mathbb{R}^{d \times d}$, $\text{head}_i \in \mathbb{R}^{b \times 1 \times \frac{d}{m}}$	$\mathbf{Y} = \text{Concat}(\text{head}_1, \dots, \text{head}_m) \mathbf{W}^o$	$O(bd^2)$	$O(2bd + d^2)$
$\mathbf{Y} \in \mathbb{R}^{b \times 1 \times d}$, $\mathbf{W}_1 \in \mathbb{R}^{d \times d_{ff}}$, $\mathbf{W}_2 \in \mathbb{R}^{d_{ff} \times d}$	Now the computations (of next token) has linear dependence on seq length.	$O(bd^2)$	$O(2bd + 8d^2)$
---	Total	$O(bd^2 + bnd)$	$O(bmn + bnd + d^2)$

Recap: complexity of decoding text

- Transformers computation for generating each word is $O(bd^2 + bnd)$.
- Remember the complexity of processing a full sequence of length: $O(bnd^2 + bn^2d)$.

Quiz: Enc-Dec inference cost

- Suppose we're using an Enc-Dec model for summarization task. Input are long passages with length m and the expected (summary length is n). The computational complexity of this operation is:
 - $O(n) + O(m)$
 - $O(n) + O(m) + O(nm)$
 - $O(n^2) + O(m^2) + O(nm)$
 - $O(n^2) + O(m^2)$

No, self attention is all-to-all and so quadratic.

Quiz: Enc-Dec inference cost

- Suppose we're using an Enc-Dec model for summarization task. Input are long passages with length m and the expected (summary length is n). The computational complexity of this operation is:
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 - $O(n^2) + O(m^2)$

No, self attention is all-to-all and so quadratic in m and n .

Quiz: Enc-Dec inference cost

- Suppose we're using an Enc-Dec model for summarization task. Input are long passages with length m and the expected (summary length is n). The computational complexity of this operation is:
 - $O(n) + O(m)$
 - $O(n) + O(m) + O(nm)$
 - $O(n^2) + O(m^2) + O(nm)$
 - $O(n^2) + O(m^2)$

No, cross attention is missing.

Quiz: Enc-Dec inference cost

- Suppose we're using an Enc-Dec model for summarization task. Input are long passages with length m and the expected (summary length is n). The computational complexity of this operation is:
 - $O(n) + O(m)$
 - $O(n) + O(m) + O(nm)$
 - $O(n^2) + O(m^2) + O(nm)$
 - $O(n^2) + O(m^2)$

Yes. The three terms are respectively the Encoder self-attention, Decoder self-attention, and Cross attention.

Though a more accurate term is: $O(n^2d + nd^2 + m^2d + md^2 + nmd)$ with d denoting the model embedding dimension.

Summary: Computational Complexity of Transformers

- **Process a sequence at once:** Computation is bounded by $O(n^2)$.
- **Processing one token at a time during inference:**
 - **KV-Cache:** To avoid redundant computations during decoding time, KV-cache is used to keep track of previous calculations of keys and values.
 - The computation is bounded by $O(n)$.
- Though in all cases, the computations are penalizable (modulo Transformer layers).
- **IO bottlenecks:** when reading/writing slows us down; we will get to this (next chapter).

Writing our own Transformer

Clone Helper Function

- Create N copies of pytorch nn.Module
- The Transformer's structure contains a lot of design repetition (like VGG)
- Remember these clones shouldn't share parameters (for the most part)

```
def clones(module, N):  
    "Produce N identical layers."  
    return nn.ModuleList([copy.deepcopy(module) for _ in range(N)])
```

Create Embedding

- Create vector representation of sequence vocabulary
- nn.Embedding creates a lookup table to map sequence vocabulary to unique vectors

```
class Embeddings(nn.Module):  
    def __init__(self, d_model, vocab):  
        super(Embeddings, self).__init__()  
        self.lut = nn.Embedding(vocab, d_model)  
        self.d_model = d_model  
  
    def forward(self, x):  
        return self.lut(x) * math.sqrt(self.d_model)
```

Positional Encoding

- Add information about an element's position in a sequence to its representation
- Element wise addition of sinusoidal encoding



```
class PositionalEncoding(nn.Module):
    "Implement the PE function."

    def __init__(self, d_model, dropout, max_len=5000):
        super(PositionalEncoding, self).__init__()
        self.dropout = nn.Dropout(p=dropout)

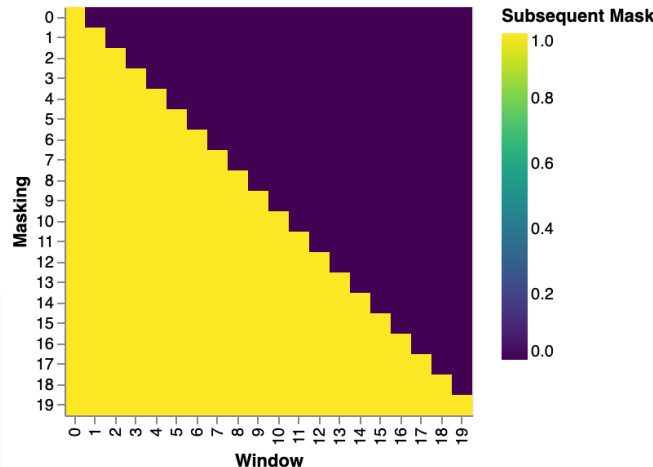
        # Compute the positional encodings once in log space.
        pe = torch.zeros(max_len, d_model)
        position = torch.arange(0, max_len).unsqueeze(1)
        div_term = torch.exp(
            torch.arange(0, d_model, 2) * -(math.log(10000.0) / d_model)
        )
        pe[:, 0::2] = torch.sin(position * div_term)
        pe[:, 1::2] = torch.cos(position * div_term)
        pe = pe.unsqueeze(0)
        self.register_buffer("pe", pe)

    def forward(self, x):
        x = x + self.pe[:, : x.size(1)].requires_grad_(False)
        return self.dropout(x)
```

Attention block

```
def attention(query, key, value, mask=None, dropout=None):  
    "Compute 'Scaled Dot Product Attention'"  
    d_k = query.size(-1)  
    scores = torch.matmul(query, key.transpose(-2, -1)) / math.sqrt(d_k)  
    if mask is not None:  
        scores = scores.masked_fill(mask == 0, -1e9)  
    p_attn = scores.softmax(dim=-1)  
    if dropout is not None:  
        p_attn = dropout(p_attn)  
    return torch.matmul(p_attn, value), p_attn
```

$-1e9$ is a large negative number,
which leads to $\text{softmax}(-1e9) \approx 0$



Multi-Head Attention

```
class MultiHeadedAttention(nn.Module):
    def __init__(self, h, d_model, dropout=0.1):
        "Take in model size and number of heads."
        super(MultiHeadedAttention, self).__init__()
        assert d_model % h == 0
        # We assume d_v always equals d_k
        self.d_k = d_model // h
        self.h = h
        self.linears = clones(nn.Linear(d_model, d_model), 4)
        self.attn = None
        self.dropout = nn.Dropout(p=dropout)
```

```
def forward(self, query, key, value, mask=None):
    "Implements Figure 2"
    if mask is not None:
        # Same mask applied to all h heads.
        mask = mask.unsqueeze(1)
        nbatches = query.size(0)

    # 1) Do all the linear projections in batch from d_model => h x d_k
    query, key, value = [
        lin(x).view(nbatches, -1, self.h, self.d_k).transpose(1, 2)
        for lin, x in zip(self.linears, (query, key, value))
    ]

    # 2) Apply attention on all the projected vectors in batch.
    x, self.attn = attention(
        query, key, value, mask=mask, dropout=self.dropout
    )

    # 3) "Concat" using a view and apply a final linear.
    x = (
        x.transpose(1, 2)
        .contiguous()
        .view(nbatches, -1, self.h * self.d_k)
    )
    del query
    del key
    del value
    return self.linears[-1](x)
```

FeedForward Layer

```
class PositionwiseFeedForward(nn.Module):
    "Implements FFN equation."

    def __init__(self, d_model, d_ff, dropout=0.1):
        super(PositionwiseFeedForward, self).__init__()
        self.w_1 = nn.Linear(d_model, d_ff)
        self.w_2 = nn.Linear(d_ff, d_model)
        self.dropout = nn.Dropout(dropout)

    def forward(self, x):
        return self.w_2(self.dropout(self.w_1(x).relu()))
```

Sublayer Connections

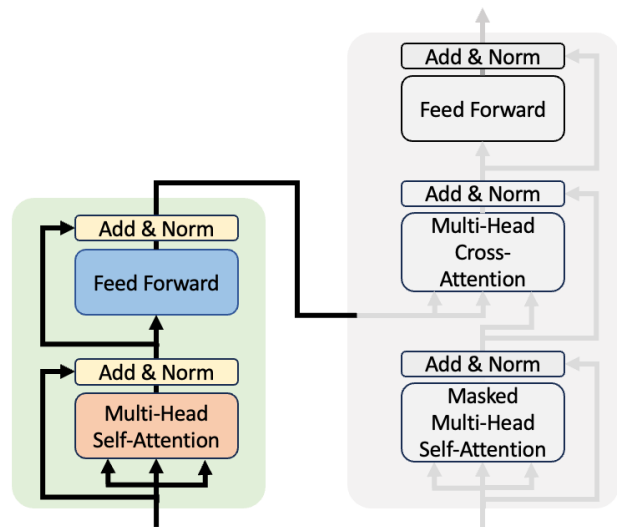
```
class SublayerConnection(nn.Module):
    """
    A residual connection followed by a layer norm.
    Note for code simplicity the norm is first as opposed to last.
    """

    def __init__(self, size, dropout):
        super(SublayerConnection, self).__init__()
        self.norm = LayerNorm(size)
        self.dropout = nn.Dropout(dropout)

    def forward(self, x, sublayer):
        "Apply residual connection to any sublayer with the same size."
        return x + self.dropout(sublayer(self.norm(x)))
```

Encoder Layer

```
class EncoderLayer(nn.Module):  
    "Encoder is made up of self-attn and feed forward (defined below)"  
  
    def __init__(self, size, self_attn, feed_forward, dropout):  
        super(EncoderLayer, self).__init__()  
        self.self_attn = self_attn  
        self.feed_forward = feed_forward  
        self.sublayer = clones(SublayerConnection(size, dropout), 2)  
        self.size = size  
  
    def forward(self, x, mask):  
        "Follow Figure 1 (left) for connections."  
        x = self.sublayer[0](x, lambda x: self.self_attn(x, x, x, mask))  
        return self.sublayer[1](x, self.feed_forward)
```



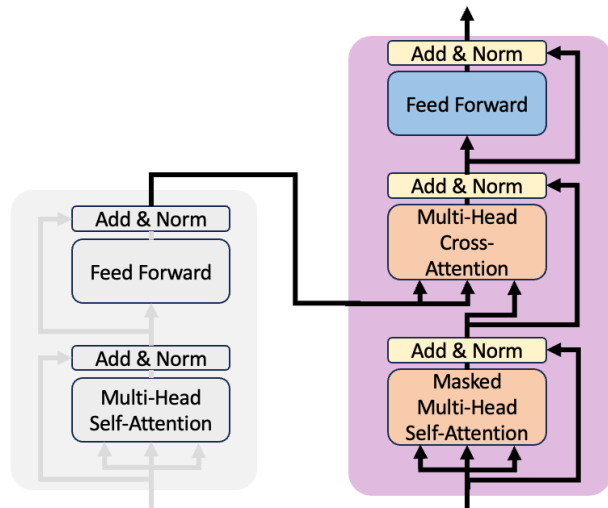
Decoder Layer

- Same as encoder layers other than:
 - the additional multi-head attention block to preform cross-attention with the output representation from the encoder

```
class DecoderLayer(nn.Module):
    "Decoder is made of self-attn, src-attn, and feed forward (defined below)"

    def __init__(self, size, self_attn, src_attn, feed_forward, dropout):
        super(DecoderLayer, self).__init__()
        self.size = size
        self.self_attn = self_attn
        self.src_attn = src_attn
        self.feed_forward = feed_forward
        self.sublayer = clones(SublayerConnection(size, dropout), 3)

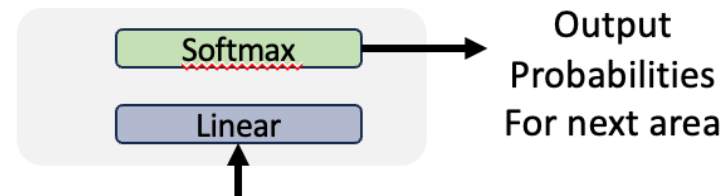
    def forward(self, x, memory, src_mask, tgt_mask):
        "Follow Figure 1 (right) for connections."
        m = memory
        x = self.sublayer[0](x, lambda x: self.self_attn(x, x, x, tgt_mask))
        x = self.sublayer[1](x, lambda x: self.src_attn(x, m, m, src_mask))
        return self.sublayer[2](x, self.feed_forward)
```



The Prediction Head

- A final linear mapping
- Apply softmax to convert logits to probabilities

```
class Generator(nn.Module):  
    "Define standard linear + softmax generation step."  
  
    def __init__(self, d_model, vocab):  
        super(Generator, self).__init__()  
        self.proj = nn.Linear(d_model, vocab)  
  
    def forward(self, x):  
        return log_softmax(self.proj(x), dim=-1)
```



Build each block

```
class Encoder(nn.Module):
    "Core encoder is a stack of N layers"

    def __init__(self, layer, N):
        super(Encoder, self).__init__()
        self.layers = clones(layer, N)
        self.norm = LayerNorm(layer.size)

    def forward(self, x, mask):
        "Pass the input (and mask) through each layer in turn."
        for layer in self.layers:
            x = layer(x, mask)
        return self.norm(x)
```

```
class Decoder(nn.Module):
    "Generic N layer decoder with masking."

    def __init__(self, layer, N):
        super(Decoder, self).__init__()
        self.layers = clones(layer, N)
        self.norm = LayerNorm(layer.size)

    def forward(self, x, memory, src_mask, tgt_mask):
        for layer in self.layers:
            x = layer(x, memory, src_mask, tgt_mask)
        return self.norm(x)
```

```

class EncoderDecoder(nn.Module):
    """
    A standard Encoder-Decoder architecture. Base for this and many
    other models.
    """

    def __init__(self, encoder, decoder, src_embed, tgt_embed, generator):
        super(EncoderDecoder, self).__init__()
        self.encoder = encoder
        self.decoder = decoder
        self.src_embed = src_embed
        self.tgt_embed = tgt_embed
        self.generator = generator

    def forward(self, src, tgt, src_mask, tgt_mask):
        "Take in and process masked src and target sequences."
        return self.decode(self.encode(src, src_mask), src_mask, tgt, tgt_mask)

    def encode(self, src, src_mask):
        return self.encoder(self.src_embed(src), src_mask)

    def decode(self, memory, src_mask, tgt, tgt_mask):
        return self.decoder(self.tgt_embed(tgt), memory, src_mask, tgt_mask)

```

```

def make_model(
    src_vocab, tgt_vocab, N=6, d_model=512, d_ff=2048, h=8, dropout=0.1
):
    "Helper: Construct a model from hyperparameters."
    c = copy.deepcopy
    attn = MultiHeadedAttention(h, d_model)
    ff = PositionwiseFeedForward(d_model, d_ff, dropout)
    position = PositionalEncoding(d_model, dropout)
    model = EncoderDecoder(
        Encoder(EncoderLayer(d_model, c(attn), c(ff), dropout), N),
        Decoder(DecoderLayer(d_model, c(attn), c(attn), c(ff), dropout), N),
        nn.Sequential(Embeddings(d_model, src_vocab), c(position)),
        nn.Sequential(Embeddings(d_model, tgt_vocab), c(position)),
        Generator(d_model, tgt_vocab),
    )

    # This was important from their code.
    # Initialize parameters with Glorot / fan_avg.
    for p in model.parameters():
        if p.dim() > 1:
            nn.init.xavier_uniform_(p)
    return model

```